

#1 · Ethical and Professional Standards

What are the six components of the CFA Institute Code of Ethics?

Answer

Members and candidates must act with integrity, competence, diligence, respect, and in an ethical manner; place the integrity of the profession and client interests above their own; use reasonable care and independent judgment; practice and encourage others to practice professionally and ethically; promote market integrity; and maintain and improve professional competence.

#2 · Ethical and Professional Standards

How many Standards of Professional Conduct are there and what are their broad categories?

Answer

There are seven Standards: Professionalism, Integrity of Capital Markets, Duties to Clients, Duties to Employers, Investment Analysis and Recommendations, Conflicts of Interest, and Responsibilities as a CFA Institute Member or Candidate.

#3 · Ethical and Professional Standards

What does Standard I(A) Knowledge of the Law require?

Answer

Members must understand and comply with applicable laws, rules, and regulations, including the Code and Standards, and must follow the stricter of applicable law or the Code and Standards when they conflict.

#4 · Ethical and Professional Standards

Under Standard I(A), what should a member do upon learning of illegal or unethical activity?

Answer

Disassociate from the activity, which may require reporting it internally or to regulators, and may include resigning from the firm if the activity continues.

#5 · Ethical and Professional Standards

What does Standard I(B) Independence and Objectivity require?

Answer

Members must use reasonable care and judgment to achieve and maintain independence and objectivity, and must not offer, solicit, or accept gifts, benefits, or compensation that could reasonably be expected to compromise their own or another's independence.

#6 · Ethical and Professional Standards

Under Standard I(B), how should gifts from clients versus issuers be treated?

Answer

Client gifts may be viewed as compensation for services if disclosed to the employer, while gifts from issuers or third parties that could threaten objectivity should be disclosed and may need to be limited to token value.

#7 · Ethical and Professional Standards

What does Standard I(C) Misrepresentation prohibit?

Answer

Members must not knowingly make any misrepresentations relating to investment analysis, recommendations, actions, or other professional activities, including omitting facts necessary to keep a statement from being misleading.

#8 · Ethical and Professional Standards

Give an example of a violation of Standard I(C) Misrepresentation.

Answer

Guaranteeing a specific investment return when no such guarantee is possible, plagiarizing another analyst's research report, or overstating one's own credentials or experience.

#9 · Ethical and Professional Standards

What does Standard I(D) Misconduct address?

Answer

Members must not engage in any professional conduct involving dishonesty, fraud, or deceit, or commit any act that reflects adversely on their professional reputation, integrity, or competence, whether or not it relates to professional activities.

#10 · Ethical and Professional Standards

Does Standard I(D) Misconduct apply only to work-related behavior?

Answer

No, it can extend to personal conduct if that conduct involves dishonesty, fraud, or deceit and reflects poorly on the member's professional reputation, integrity, or competence.

#11 · Ethical and Professional Standards

What is Standard II(A) Material Nonpublic Information?

Answer

Members who possess material nonpublic information that could affect an investment's value must not act or cause others to act on it, and must make reasonable efforts to prevent its misuse.

#12 · Ethical and Professional Standards

What is the mosaic theory under Standard II(A)?

Answer

Combining public information and non-material nonpublic information to reach an investment conclusion is permitted, even if the resulting conclusion would be the same as if material nonpublic information had been used.

#13 · Ethical and Professional Standards

What firewall procedures help comply with Standard II(A)?

Answer

Firms often use information barriers between departments (such as investment banking and research), restricted lists, watch lists, and physical separation of personnel to prevent misuse of material nonpublic information.

#14 · Ethical and Professional Standards

What does Standard II(B) Market Manipulation prohibit?

Answer

Members must not engage in practices that distort prices or artificially inflate trading volume with the intent to mislead market participants, including both information-based and transaction-based manipulation.

#15 · Ethical and Professional Standards

Give an example of transaction-based market manipulation under Standard II(B).

Answer

Wash trading, painting the tape by trading among related accounts to create the appearance of activity, or entering orders with no intention of executing them to move the market price.

#16 · Ethical and Professional Standards

What does Standard III(A) Loyalty, Prudence, and Care require?

Answer

Members must act for the benefit of clients and place client interests before their own or their employer's, exercising prudence, care, skill, and diligence in managing client accounts.

#17 · Ethical and Professional Standards

How does the duty of loyalty under Standard III(A) apply to soft dollar or brokerage arrangements?

Answer

Client brokerage is client property and must be used to benefit the client, such as through client-directed brokerage that still seeks best execution, not to benefit the manager alone.

#18 · Ethical and Professional Standards

What does Standard III(B) Fair Dealing require?

Answer

Members must deal fairly and objectively with all clients when providing investment analysis, recommendations, or taking investment action, and must not favor some clients over others.

#19 · Ethical and Professional Standards

Under Standard III(B), how should an investment recommendation change be disseminated?

Answer

Simultaneously or as close to simultaneously as possible to all clients who are entitled to receive it, rather than giving preferential early access to select clients.

#20 · Ethical and Professional Standards

What does Standard III(C) Suitability require?

Answer

Members must make a reasonable inquiry into a client's investment objectives, risk tolerance, time horizon, and financial constraints before making a recommendation, and must consider a client's whole portfolio when giving advice.

#21 · Ethical and Professional Standards

How does Standard III(C) Suitability differ for pooled investment vehicles?

Answer

For a fund managed to a stated mandate, the manager must ensure investment actions are consistent with the fund's stated objectives and policies rather than any individual investor's personal circumstances.

#22 · Ethical and Professional Standards

What does Standard III(D) Performance Presentation require?

Answer

When communicating investment performance, members must make reasonable efforts to ensure it is fair, accurate, and complete, avoiding cherry-picked time periods or misleading benchmarks.

#23 · Ethical and Professional Standards

What does Standard III(E) Preservation of Confidentiality require?

Answer

Members must keep information about current, former, and prospective clients confidential unless disclosure is required by law, the client permits it, or the information concerns illegal activity by the client.

#24 · Ethical and Professional Standards

Does Standard III(E) confidentiality survive the end of a client relationship?

Answer

Yes, the duty of confidentiality continues even after the client relationship has ended.

#25 · Ethical and Professional Standards

What does Standard IV(A) Loyalty require regarding an employer?

Answer

Members must act for the benefit of their employer and not deprive the employer of their skills and abilities, divulge confidential information, or otherwise cause harm to the employer.

#26 · Ethical and Professional Standards

Under Standard IV(A), what preparations are allowed before leaving an employer?

Answer

A member may make arrangements to compete with a current employer after departure, such as lining up office space or logistics, but must not solicit clients, misappropriate confidential records, or otherwise misuse employer resources before leaving.

#27 · Ethical and Professional Standards

What does Standard IV(B) Additional Compensation Arrangements require?

Answer

Members must not accept gifts, benefits, or compensation that competes with or might reasonably be expected to create a conflict with their employer's interest, unless they obtain written consent from all parties involved.

#28 · Ethical and Professional Standards

What does Standard IV(C) Responsibilities of Supervisors require?

Answer

Supervisors must make reasonable efforts to detect and prevent violations of applicable laws, rules, and the Code and Standards by anyone subject to their supervision or authority.

#29 · Ethical and Professional Standards

What is an adequate compliance system under Standard IV(C)?

Answer

One that is reasonably designed to detect and prevent violations, with clear procedures communicated to employees, and is actively enforced rather than just existing on paper.

#30 · Ethical and Professional Standards

What does Standard V(A) Diligence and Reasonable Basis require?

Answer

Members must exercise diligence, independence, and thoroughness in analyzing investments, making recommendations, and taking investment actions, and must have a reasonable and adequate basis supported by appropriate research.

#31 · Ethical and Professional Standards

Under Standard V(A), what if a member relies on third-party research?

Answer

The member should establish a policy to periodically evaluate whether that third-party research has a sound basis, and should not simply accept it without any due diligence.

#32 · Ethical and Professional Standards

What does Standard V(B) Communication with Clients and Prospective Clients require?

Answer

Members must disclose the basic format and general principles of investment processes used, promptly disclose significant changes affecting those processes, and distinguish between fact and opinion when making recommendations.

#33 · Ethical and Professional Standards

Under Standard V(B), what must members disclose about investment risk?

Answer

They must communicate the risks and limitations of the analysis or recommendation, including any significant assumptions or limitations of the investment process.

#34 · Ethical and Professional Standards

What does Standard V(C) Record Retention require?

Answer

Members must develop and maintain records supporting their investment analysis, recommendations, actions, and other investment-related communications, and such records are generally considered the property of the firm.

#35 · Ethical and Professional Standards

What is the typical record retention guideline referenced under Standard V(C) absent other regulation?

Answer

CFA Institute recommends maintaining records for at least seven years in the absence of any other regulatory or firm requirement.

#36 · Ethical and Professional Standards

What does Standard VI(A) Disclosure of Conflicts require?

Answer

Members must make full and fair disclosure of all matters that could reasonably be expected to impair their independence and objectivity or interfere with duties to clients, employers, and prospective clients.

#37 · Ethical and Professional Standards

Give an example of a conflict that must be disclosed under Standard VI(A).

Answer

A member's ownership of stock in a company they are recommending, board membership at an issuer being covered, or a referral fee arrangement between the firm and another party.

#38 · Ethical and Professional Standards

What does Standard VI(B) Priority of Transactions require?

Answer

Investment transactions for clients and employers must have priority over transactions in securities or investments in which a member is the beneficial owner, so client interests come first.

#39 · Ethical and Professional Standards

Under Standard VI(B), what is generally required before a member trades personally in a security also being recommended?

Answer

Clients and the employer should have an adequate opportunity to act on a recommendation before the member trades for their own personal account in the same security.

#40 · Ethical and Professional Standards

What does Standard VI(C) Referral Fees require?

Answer

Members must disclose to employers, clients, and prospective clients any compensation, consideration, or benefit received from, or paid to, others for the recommendation of products or services.

#41 · Ethical and Professional Standards

What does Standard VII(A) Conduct as Members and Candidates in the CFA Program prohibit?

Answer

Members and candidates must not engage in conduct that compromises the reputation or integrity of CFA Institute or the CFA designation, such as cheating on the exam, revealing confidential exam content, or misusing CFA Institute materials.

#42 · Ethical and Professional Standards

What does Standard VII(B) Reference to CFA Institute, the CFA Designation, and the CFA Program require?

Answer

Members and candidates must not misrepresent or exaggerate the meaning or implications of CFA Institute membership, holding the CFA designation, or candidacy in the CFA Program.

#43 · Ethical and Professional Standards

Can a candidate say they are a 'CFA Level II candidate' before completing Level I?

Answer

No, a candidate may only reference an exam level once they have registered for and are eligible to sit for that specific level, and must not imply a superior status is guaranteed by candidacy alone.

#44 · Ethical and Professional Standards

What are the Global Investment Performance Standards (GIPS)?

Answer

GIPS are a set of voluntary, ethical standards for calculating and presenting investment performance based on the principles of fair representation and full disclosure, promoting comparability across firms and time periods.

#45 · Ethical and Professional Standards

Why might an investment firm choose to claim compliance with GIPS?

Answer

Claiming GIPS compliance signals credibility and comparability to prospective clients, since it demonstrates the firm calculates and presents composite performance using consistent, industry-accepted, verifiable methodology.

#46 · Ethical and Professional Standards

What are the two main components of the CFA Institute Code and Standards?

Answer

The Code of Ethics, which states broad ethical principles members and candidates agree to uphold, and the Standards of Professional Conduct, which give specific, enforceable guidance for applying those principles.

#47 · Ethical and Professional Standards

A portfolio manager uses confidential client information from a former employer at her new firm. Which Standard is violated?

Answer

Standard I(A), Knowledge of the Law, and Standard IV(A), Loyalty, are implicated, since using a former employer's confidential records without consent breaches duties owed to that employer even after departure.

#48 · Ethical and Professional Standards

Standard I(B): Independence and Objectivity, what does it require?

Answer

Members must use reasonable care and judgment to achieve and maintain independence and objectivity, avoiding gifts, favors, or pressures from third parties that could compromise their professional judgment.

#49 · Ethical and Professional Standards

An analyst accepts a lavish trip from a company he covers, then issues a buy rating. Which Standard applies?

Answer

Standard I(B), Independence and Objectivity, is violated because the gift could reasonably be seen as compromising the analyst's objectivity, regardless of whether the rating itself was actually biased.

#50 · Ethical and Professional Standards

What does Standard I(C), Misrepresentation, prohibit?

Answer

Members must not knowingly make false or misleading statements about investment analysis, recommendations, actions, or other professional activities, including omitting facts necessary to keep a statement from being misleading.

#51 · Ethical and Professional Standards

A firm's marketing claims it 'guarantees' a fund will beat its benchmark. Which Standard is violated?

Answer

Standard I(C), Misrepresentation, since guaranteeing investment performance is a false and misleading claim that cannot be substantiated.

#52 · Ethical and Professional Standards

What does Standard I(D), Misconduct, cover?

Answer

Members must not engage in professional conduct involving dishonesty, fraud, or deceit, or commit any act that reflects adversely on their professional reputation, integrity, or competence, including in personal (non-work) matters.

#53 · Ethical and Professional Standards

An analyst is convicted of tax fraud unrelated to his job duties. Does this violate the Code and Standards?

Answer

Yes, Standard I(D), Misconduct, applies because the conviction involves dishonesty and reflects adversely on the member's professional reputation and integrity, even though it is unrelated to his investment work.

#54 · Ethical and Professional Standards

What is required under Standard II(A), Material Nonpublic Information?

Answer

Members who possess material nonpublic information must not act or induce others to act on it; they must not trade on or disclose such information to others.

#55 · Ethical and Professional Standards

What is the 'mosaic theory' and why is it relevant to Standard II(A)?

Answer

The mosaic theory holds that an analyst may combine public information and non-material nonpublic information to form an investment conclusion without violating Standard II(A), because no single material nonpublic fact drove the conclusion.

#56 · Ethical and Professional Standards

What does Standard II(B), Market Manipulation, prohibit?

Answer

Members must not engage in practices that distort prices or artificially inflate trading volume, including both information-based manipulation (spreading false rumors) and transaction-based manipulation (wash trades).

#57 · Ethical and Professional Standards

A trader places and cancels large orders to create a false impression of demand, then trades on the resulting price move. Which Standard is violated?

Answer

Standard II(B), Market Manipulation, since this is transaction-based manipulation designed to distort the security's price or trading volume.

#58 · Ethical and Professional Standards

List the five subsections under Standard III, Duties to Clients.

Answer

III(A) Loyalty, Prudence, and Care; III(B) Fair Dealing; III(C) Suitability; III(D) Performance Presentation; III(E) Preservation of Confidentiality.

#59 · Ethical and Professional Standards

What does Standard III(A), Loyalty, Prudence, and Care, require?

Answer

Members must act with reasonable care and exercise prudent judgment, acting for the benefit of clients and placing client interests before their employer's or their own interests.

#60 · Ethical and Professional Standards

What does Standard III(B), Fair Dealing, require regarding trade allocation?

Answer

Members must deal fairly and objectively with all clients when disseminating investment recommendations or taking investment action, such as allocating trades and IPO shares fairly rather than favoring certain clients.

#61 · Ethical and Professional Standards

A manager tells his favorite client about a rating downgrade an hour before informing other clients. Which Standard is violated?

Answer

Standard III(B), Fair Dealing, because clients must be treated fairly when investment recommendations are disseminated, not selectively favored by relationship.

#62 · Ethical and Professional Standards

What must a member do under Standard III(C), Suitability, before investing on a client's behalf?

Answer

The member must make a reasonable inquiry into the client's investment experience, risk tolerance, objectives, and constraints, then judge each investment's suitability within the context of the client's total portfolio.

#63 · Ethical and Professional Standards

A conservative retiree is placed heavily into speculative leveraged options without any documented risk assessment. Which Standard is violated?

Answer

Standard III(C), Suitability, since the investment does not match the client's risk tolerance and objectives and no reasonable inquiry into those factors was documented.

#64 · Ethical and Professional Standards

What does Standard III(D), Performance Presentation, require?

Answer

When communicating investment performance, members must make reasonable efforts to ensure it is fair, accurate, and complete, avoiding cherry-picked time periods or misleading comparisons.

#65 · Ethical and Professional Standards

A firm advertises only its best-performing fund's returns from a cherry-picked 3-month window. Which Standard is violated?

Answer

Standard III(D), Performance Presentation, because the presentation is not fair, accurate, or complete and could mislead prospective clients about typical performance.

#66 · Ethical and Professional Standards

What does Standard III(E), Preservation of Confidentiality, require?

Answer

Members must keep current, former, and prospective client information confidential unless disclosure is required by law, the client consents, or the information concerns illegal activity by the client.

#67 · Ethical and Professional Standards

A client reveals during a meeting that he is running a Ponzi scheme with the assets under management. May the member disclose this?

Answer

Yes, Standard III(E) permits disclosure of confidential client information when it concerns illegal activities by the client, even without the client's consent.

#68 · Ethical and Professional Standards

List the three subsections under Standard IV, Duties to Employers.

Answer

IV(A) Loyalty; IV(B) Additional Compensation Arrangements; IV(C) Responsibilities of Supervisors.

#69 · Ethical and Professional Standards

What does Standard IV(A), Loyalty, require regarding an employer's business?

Answer

Members must act for the benefit of their employer and not deprive the employer of their skills, divulge confidential information, or otherwise cause harm, though they may make certain arrangements to prepare for independent competitive practice before leaving.

#70 · Ethical and Professional Standards

May an analyst plan to start a competing firm while still employed, under Standard IV(A)?

Answer

Yes, undertaking preliminary arrangements such as drafting a business plan or arranging financing is generally permitted, but soliciting current clients or misappropriating confidential records before departure is not.

#71 · Ethical and Professional Standards

What does Standard IV(B), Additional Compensation Arrangements, require?

Answer

Members must not accept gifts, benefits, or compensation from clients or others that could create a conflict of interest with their employer's interests, unless they obtain written consent from all parties involved.

#72 · Ethical and Professional Standards

A client offers an advisor a bonus for good performance, separate from the employer's compensation plan, without telling the firm. Which Standard is violated?

Answer

Standard IV(B), Additional Compensation Arrangements, because the advisor must obtain written consent from the employer before accepting outside compensation that could create a conflict of interest.

#73 · Ethical and Professional Standards

What does Standard IV(C), Responsibilities of Supervisors, require?

Answer

Members with supervisory authority must make reasonable efforts to detect and prevent violations of laws, regulations, and the Code and Standards by those under their supervision, including establishing an adequate compliance system.

#74 · Ethical and Professional Standards

A compliance officer ignores repeated red flags of unauthorized trading by a subordinate. Which Standard is violated?

Answer

Standard IV(C), Responsibilities of Supervisors, because the supervisor failed to make reasonable efforts to detect and prevent a known or suspected violation.

#75 · Ethical and Professional Standards

List the three subsections under Standard V, Investment Analysis, Recommendations, and Actions.

Answer

V(A) Diligence and Reasonable Basis; V(B) Communication with Clients and Prospective Clients; V(C) Record Retention.

#76 · Ethical and Professional Standards

What does Standard V(A), Diligence and Reasonable Basis, require?

Answer

Members must exercise diligence, independence, and thoroughness in analyzing investments, making recommendations, or taking investment actions, and have a reasonable and adequate basis supported by appropriate research.

#77 · Ethical and Professional Standards

An analyst issues a buy recommendation based solely on an unverified tip from a friend, with no independent research. Which Standard is violated?

Answer

Standard V(A), Diligence and Reasonable Basis, since the recommendation lacks a reasonable and adequate basis grounded in appropriate research.

#78 · Ethical and Professional Standards

What must members disclose under Standard V(B), Communication with Clients and Prospective Clients?

Answer

The basic format and general principles of the investment process used, any changes that might materially affect that process, and the factors important to the investment recommendation, while distinguishing fact from opinion.

#79 · Ethical and Professional Standards

How long must members retain records supporting investment analyses under Standard V(C), Record Retention, if local law is silent?

Answer

CFA Institute recommends retaining records for at least seven years in the absence of applicable regulatory or firm requirements.

#80 · Ethical and Professional Standards

List the three subsections under Standard VI, Conflicts of Interest.

Answer

VI(A) Disclosure of Conflicts; VI(B) Priority of Transactions; VI(C) Referral Fees.

#81 · Ethical and Professional Standards

What does Standard VI(A), Disclosure of Conflicts, require?

Answer

Members must make full and fair disclosure of all matters that could reasonably be expected to impair their independence or objectivity or interfere with duties to clients, employers, and prospective clients.

#82 · Ethical and Professional Standards

An advisor recommends a fund managed by his firm's affiliate but never mentions the affiliation to the client. Which Standard is violated?

Answer

Standard VI(A), Disclosure of Conflicts, because the affiliation is a conflict of interest that must be fully and fairly disclosed to the client.

#83 · Ethical and Professional Standards

What does Standard VI(B), Priority of Transactions, require?

Answer

Client and employer transactions must take priority over transactions in securities where a member is the beneficial owner, so personal trades should not disadvantage clients, such as by front-running.

#84 · Ethical and Professional Standards

A portfolio manager buys a stock for his personal account the day before placing a large buy order for clients. Which Standard is violated?

Answer

Standard VI(B), Priority of Transactions, since this is front-running that gives the manager's personal trade priority and potential advantage over client transactions.

#85 · Ethical and Professional Standards

What does Standard VI(C), Referral Fees, require?

Answer

Members must disclose to employers, clients, and prospective clients any compensation or benefit received from, or paid to, others for the recommendation of products or services.

#86 · Ethical and Professional Standards

What does Standard VII(A), Conduct as Members and Candidates in the CFA Program, prohibit?

Answer

Conduct that compromises the reputation or integrity of CFA Institute, the CFA designation, or the CFA examinations, such as cheating on the exam, revealing confidential exam content, or misusing Institute materials.

#87 · Ethical and Professional Standards

A candidate posts specific exam questions online right after sitting the CFA exam. Which Standard is violated?

Answer

Standard VII(A), because disclosing confidential exam content breaks the candidate's agreement with CFA Institute and compromises the integrity of the CFA Program.

#88 · Ethical and Professional Standards

What does Standard VII(B), Reference to CFA Institute, the CFA Designation, and the CFA Program, require?

Answer

When referencing CFA Institute membership, the CFA designation, or candidacy, members and candidates must not misrepresent or exaggerate the meaning or implications of holding the credential or being a candidate.

#89 · Ethical and Professional Standards

A Level I candidate lists 'CFA' after her name on her business card before earning the charter. Which Standard is violated?

Answer

Standard VII(B), Reference to CFA Institute, the CFA Designation, and the CFA Program, since only charterholders who have completed all three levels and met experience requirements may use the CFA designation.

#90 · Ethical and Professional Standards

A firm claims GIPS compliance but excludes its worst-performing composite from all reporting. What is wrong under GIPS and the Standards?

Answer

This violates the GIPS requirement that firms include all actual, fee-paying, discretionary portfolios in at least one composite, and it also implicates Standard I(C), Misrepresentation, since selectively excluding poor performance misleads prospective clients about the firm's true track record.

#91 · Quantitative Methods

Future value formula (single sum)

Answer

$FV = PV \times (1 + r)^N$, where PV is the present value, r is the interest rate per period, and N is the number of compounding periods.

#92 · Quantitative Methods

Present value formula (single sum)

Answer

$PV = FV \div (1 + r)^N$. This discounts a future cash flow back to today at rate r over N periods.

#93 · Quantitative Methods

Effective annual rate (EAR) formula

Answer

$EAR = (1 + \text{periodic rate})^m - 1$, where m is the number of compounding periods per year. EAR lets rates with different compounding frequencies be compared fairly.

#94 · Quantitative Methods

Continuous compounding EAR formula

Answer

$EAR = e^{(\text{stated annual rate})} - 1$. As compounding frequency approaches infinity, this is the limiting case of the EAR formula.

#95 · Quantitative Methods

Future value of an ordinary annuity

Answer

$FV = A \times [((1 + r)^N - 1) \div r]$, where A is the periodic payment, r is the rate per period, and N is the number of payments made at the END of each period.

#96 · Quantitative Methods

Present value of an ordinary annuity

Answer

$PV = A \times [(1 - (1 + r)^{-N}) \div r]$, where A is the periodic payment. Use this to value a level stream of payments starting one period from today.

#97 · Quantitative Methods

Annuity due vs ordinary annuity

Answer

An annuity due has payments at the BEGINNING of each period. Its value equals the ordinary annuity value $\times (1 + r)$, since every cash flow is shifted one period earlier.

#98 · Quantitative Methods

Present value of a perpetuity

Answer

$PV = A \div r$, where A is the constant periodic payment and r is the discount rate per period. A perpetuity is a level cash flow stream with no end date.

#99 · Quantitative Methods

Net present value (NPV)

Answer

NPV = sum of the present values of all expected cash flows minus the initial outlay. A project with $NPV > 0$ is expected to increase firm value and should generally be accepted.

#100 · Quantitative Methods

Internal rate of return (IRR)

Answer

IRR is the discount rate that makes NPV equal to zero, that is, the rate at which the present value of inflows equals the present value of outflows.

#101 · Quantitative Methods

Holding period return (HPR)

Answer

$$\text{HPR} = (\text{Ending value} - \text{Beginning value} + \text{Income}) \div \text{Beginning value}$$

It measures total return earned over the actual period the investment was held.

#102 · Quantitative Methods

Arithmetic mean return

Answer

The simple average of a series of periodic returns, calculated as the sum of returns divided by the number of periods. It does not account for compounding.

#103 · Quantitative Methods

Geometric mean return

Answer

Geometric mean = $[(1+R_1) \times (1+R_2) \times \dots \times (1+R_n)]^{(1 \div n)} - 1$. It reflects the actual compound growth rate over multiple periods and is always $\leq$ the arithmetic mean when returns vary.

#104 · Quantitative Methods

Money-weighted rate of return

Answer

The internal rate of return on a portfolio, which accounts for the size and timing of all cash flows (contributions and withdrawals), effectively an IRR applied to investment cash flows.

#105 · Quantitative Methods

Time-weighted rate of return

Answer

The compound growth rate of one unit of currency invested in a portfolio, calculated by linking sub-period returns and removing the effect of external cash flow timing. Preferred for evaluating a manager's skill.

#106 · Quantitative Methods

Population variance formula

Answer

$\sigma^2 = \Sigma(X_i - \mu)^2 \div N$, where μ is the population mean and N is the number of observations in the population.

#107 · Quantitative Methods

Sample variance formula

Answer

$s^2 = \Sigma(X_i - \bar{X})^2 \div (n - 1)$, where $\bar{X}$ is the sample mean and $n - 1$ is the degrees of freedom, used because it produces an unbiased estimator of the population variance.

#108 · Quantitative Methods

Standard deviation

Answer

The positive square root of variance, $s = \sqrt{s^2}$. It measures dispersion in the same units as the original data, unlike variance which is in squared units.

#109 · Quantitative Methods

Coefficient of variation (CV)

Answer

$CV = s \div \bar{X}$, the standard deviation divided by the mean. It expresses risk per unit of expected return, allowing comparison of dispersion across data sets with different means.

#110 · Quantitative Methods

Sharpe ratio

Answer

Sharpe ratio = $(R_p - R_f) \div \sigma_p$, where R_p is portfolio return, R_f is the risk-free rate, and σ_p is the portfolio's standard deviation. It measures excess return earned per unit of total risk.

#111 · Quantitative Methods

Covariance

Answer

$\text{Cov}(X, Y) = \Sigma[(X_i - \bar{X})(Y_i - \bar{Y})] \div (n - 1)$. It measures how two variables move together; positive values mean they tend to move in the same direction.

#112 · Quantitative Methods

Correlation coefficient

Answer

$r = \text{Cov}(X, Y) \div (\sigma_X \times \sigma_Y)$. It rescales covariance to range between -1 and $+1$, showing both the direction and strength of a linear relationship between two variables.

#113 · Quantitative Methods

Skewness

Answer

Skewness measures the asymmetry of a distribution. Positive skew has a long right tail (mean > median > mode), negative skew has a long left tail (mean < median < mode), and a symmetric distribution has skewness of zero.

#114 · Quantitative Methods

Kurtosis

Answer

Kurtosis measures the peakedness of a distribution and the thickness of its tails relative to a normal distribution. A normal distribution has excess kurtosis of 0; leptokurtic distributions (excess kurtosis > 0) have fatter tails and more peakedness.

#115 · Quantitative Methods

Chebyshev's inequality

Answer

For any distribution, the proportion of observations within k standard deviations of the mean is at least $1 - 1/k^2$, for $k > 1$. It applies regardless of the shape of the distribution.

#116 · Quantitative Methods

z-score formula

Answer

$z = (X - \mu) \div \sigma$. It standardizes an observation by expressing how many standard deviations it lies from the mean, allowing comparison across different distributions.

#117 · Quantitative Methods

Properties of the normal distribution

Answer

The normal distribution is symmetric and bell-shaped, fully described by its mean and variance, has skewness of 0 and kurtosis of 3, and about 68%, 95%, and 99% of observations fall within 1, 2, and 3 standard deviations of the mean respectively.

#118 · Quantitative Methods

Empirical rule (normal distribution)

Answer

Approximately 68% of observations lie within $\pm 1\sigma$ of the mean, approximately 95% lie within $\pm 2\sigma$, and approximately 99% lie within $\pm 3\sigma$.

#119 · Quantitative Methods

Multiplication rule for independent events

Answer

$P(A \text{ and } B) = P(A) \times P(B)$ when events A and B are independent, meaning the occurrence of one does not affect the probability of the other.

#120 · Quantitative Methods

Addition rule for probability

Answer

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. Subtracting the joint probability avoids double-counting the overlap between the two events.

#121 · Quantitative Methods

Conditional probability formula

Answer

$P(A | B) = P(A \text{ and } B) \div P(B)$, the probability of event A occurring given that event B has already occurred, valid when $P(B) > 0$.

#122 · Quantitative Methods

Bayes' formula

Answer

$P(A | B) = [P(B | A) \div P(B)] \times P(A)$. It updates the prior probability of A using new information from event B to produce a posterior probability.

#123 · Quantitative Methods

Expected value of a random variable

Answer

$E(X) = \sum [P(X_i) \times X_i]$, the probability-weighted average of all possible outcomes of a random variable.

#124 · Quantitative Methods

Central Limit Theorem

Answer

For a sufficiently large sample size (typically $n \geq 30$), the sampling distribution of the sample mean approaches a normal distribution regardless of the shape of the underlying population distribution.

#125 · Quantitative Methods

Standard error of the sample mean

Answer

When the population standard deviation σ is known, the standard error = $\sigma \div \sqrt{n}$. When σ is unknown, it is estimated using the sample standard deviation $s \div \sqrt{n}$.

#126 · Economics

What is the law of demand?

Answer

Holding all else constant, the quantity demanded of a good falls as its price rises, and rises as its price falls, producing a downward-sloping demand curve.

#127 · Economics

What is the law of supply?

Answer

Holding all else constant, the quantity supplied of a good rises as its price rises, and falls as its price falls, producing an upward-sloping supply curve.

#128 · Economics

Define own-price elasticity of demand.

Answer

The percentage change in quantity demanded divided by the percentage change in the good's own price. It measures how responsive buyers are to a price change.

#129 · Economics

What does it mean when demand is price inelastic?

Answer

The absolute value of own-price elasticity is less than 1, so a given percentage price change produces a smaller percentage change in quantity demanded, and total expenditure moves in the same direction as price.

#130 · Economics

What is cross-price elasticity of demand and what sign indicates substitutes?

Answer

It measures the percentage change in quantity demanded of one good relative to the percentage change in the price of another good. A positive cross-price elasticity indicates the two goods are substitutes.

#131 • Economics

What is income elasticity of demand and how are normal versus inferior goods distinguished?

Answer

It measures the percentage change in quantity demanded relative to the percentage change in consumer income. Normal goods have positive income elasticity, while inferior goods have negative income elasticity.

#132 • Economics

What is consumer surplus?

Answer

The difference between the maximum price a consumer is willing to pay for each unit and the price actually paid, summed across all units purchased, represented as the area below the demand curve and above the market price.

#133 · Economics

What is producer surplus?

Answer

The difference between the price a producer receives and the minimum price it would accept to supply each unit, represented as the area above the supply curve and below the market price.

#134 · Economics

What condition defines allocative efficiency in a competitive market?

Answer

Allocative efficiency occurs where marginal benefit equals marginal cost, which happens at the equilibrium quantity where supply and demand curves intersect, maximizing total surplus.

#135 · Economics

What is a deadweight loss?

Answer

The reduction in total economic surplus that results when a market operates away from the competitive equilibrium quantity, such as from a price ceiling, price floor, tax, or monopoly restriction of output.

#136 · Economics

What are the defining characteristics of perfect competition?

Answer

Many small firms selling a homogeneous product, free entry and exit, perfect information, and each firm acting as a price taker with a perfectly elastic (horizontal) demand curve.

#137 · Economics

In perfect competition, what rule determines a firm's profit-maximizing output in the short run?

Answer

A firm maximizes profit by producing the quantity at which marginal revenue equals marginal cost, and it will continue to operate in the short run as long as price covers average variable cost.

#138 · Economics

What happens to economic profit for firms in perfect competition in the long run?

Answer

Free entry and exit drive economic profit to zero, as new firms enter when profits are positive and existing firms exit when profits are negative, until price equals minimum long-run average total cost.

#139 · Economics

What are the key features of a monopoly market structure?

Answer

A single seller of a product with no close substitutes, high barriers to entry, and the firm faces the entire downward-sloping market demand curve, giving it pricing power.

#140 · Economics

How does a profit-maximizing monopolist set price relative to marginal cost?

Answer

The monopolist produces where marginal revenue equals marginal cost, then charges the price on the demand curve at that quantity, which is above marginal cost, unlike the perfectly competitive outcome.

#141 · Economics

What distinguishes monopolistic competition from perfect competition?

Answer

Monopolistic competition has many firms and free entry and exit like perfect competition, but each firm sells a differentiated product, giving it some pricing power and a downward-sloping demand curve.

#142 · Economics

What is the defining feature of an oligopoly market structure?

Answer

A small number of large firms dominate the market, each firm's decisions materially affect its rivals, and firms are mutually interdependent, often leading to strategic behavior modeled with game theory.

#143 · Economics

Define gross domestic product (GDP).

Answer

The market value of all final goods and services produced within an economy's borders during a specific period, whether by domestic or foreign-owned factors of production.

#144 · Economics

What is the expenditure approach to calculating GDP?

Answer

GDP equals consumption plus gross investment plus government spending plus net exports (exports minus imports), summing all final spending on domestically produced goods and services.

#145 · Economics

What is the income approach to calculating GDP?

Answer

GDP is calculated by summing all factor incomes earned in production, such as compensation of employees, rent, interest, and profits, plus statistical adjustments like depreciation and net indirect taxes.

#146 · Economics

What is the difference between nominal GDP and real GDP?

Answer

Nominal GDP measures output at current-year prices, while real GDP measures output using constant base-year prices, so real GDP removes the effect of price changes and isolates true output growth.

#147 · Economics

What are the phases of the business cycle?

Answer

Expansion, peak, contraction (recession), and trough, describing recurring fluctuations in aggregate economic activity around a long-term growth trend.

#148 · Economics

What defines a recession in common usage?

Answer

A significant decline in economic activity spread across the economy, commonly approximated as at least two consecutive quarters of negative real GDP growth, along with falling employment and income.

#149 · Economics

What is the difference between leading, coincident, and lagging economic indicators?

Answer

Leading indicators change before the economy turns, coincident indicators move at the same time as overall economic activity, and lagging indicators change after the economy has already shifted direction.

#150 · Economics

How is the unemployment rate calculated?

Answer

The unemployment rate equals the number of unemployed persons divided by the labor force (employed plus unemployed persons actively seeking work), expressed as a percentage.

#151 · Economics

What is the difference between the Consumer Price Index (CPI) and the GDP deflator as inflation measures?

Answer

CPI tracks the cost of a fixed basket of goods and services purchased by a typical household, while the GDP deflator reflects prices of all domestically produced goods and services and uses a basket that changes with current output.

#152 · Economics

Distinguish demand-pull inflation from cost-push inflation.

Answer

Demand-pull inflation arises when aggregate demand grows faster than the economy's productive capacity, while cost-push inflation arises when rising input costs, such as wages or raw materials, push up prices even as demand is stable.

#153 · Economics

What are the primary tools of monetary policy used by a central bank?

Answer

Adjusting the policy interest rate, conducting open market operations to influence the money supply, and setting reserve requirements for commercial banks, all aimed at achieving price stability and supporting output.

#154 · Economics

What is expansionary fiscal policy and what are its main tools?

Answer

Fiscal policy aimed at stimulating aggregate demand and economic growth, implemented through increased government spending and/or reduced taxes, typically financed by higher government borrowing.

#155 · Economics

What is the quantity theory of money?

Answer

The equation $MV = PY$, where the money supply times velocity of money equals the price level times real output, implying that in the long run, sustained growth in the money supply beyond real output growth causes inflation.

#156 · Economics

What is purchasing power parity (PPP) in the context of exchange rates?

Answer

The theory that exchange rates should adjust so that identical goods cost the same when prices are converted into a common currency, meaning currencies of countries with higher inflation should depreciate over time relative to lower-inflation countries.

#157 · Economics

What is covered interest rate parity?

Answer

A no-arbitrage condition stating that the forward exchange rate premium or discount between two currencies must offset the interest rate differential between the two countries, so hedged returns are equal across currencies.

#158 · Economics

What does it mean for a currency to appreciate or depreciate under a floating exchange rate regime?

Answer

Appreciation means a currency's value rises relative to another currency, so fewer units are needed to buy one unit of the other currency, while depreciation means its value falls and more units are needed, both driven by market supply and demand.

#159 · Economics

What is the difference between a current account and a capital/financial account in the balance of payments?

Answer

The current account records trade in goods and services, income flows, and current transfers, while the capital and financial account records cross-border transactions in financial assets and capital transfers; the two must sum to balance overall payments.

#160 · Economics

What is comparative advantage and why does it drive gains from trade?

Answer

A country has comparative advantage in producing a good if it can produce it at a lower opportunity cost than another country. Trade based on comparative advantage allows both countries to consume more than they could in isolation, even if one country is absolutely more efficient at producing everything.

#161 · Financial Statement Analysis

What is the core principle of revenue recognition under IFRS 15 and ASC 606?

Answer

Revenue is recognized when a company satisfies a performance obligation by transferring control of a good or service to a customer, in an amount reflecting the consideration the company expects to receive.

#162 · Financial Statement Analysis

What is the five-step model for revenue recognition under the converged standard?

Answer

1) Identify the contract with a customer, 2) identify the separate performance obligations, 3) determine the transaction price, 4) allocate the transaction price to each obligation, 5) recognize revenue when or as each obligation is satisfied.

#163 · Financial Statement Analysis

How does revenue differ from a gain on the income statement?

Answer

Revenue arises from a company's ordinary, ongoing business activities such as selling goods or services. A gain arises from peripheral or incidental transactions, such as selling a piece of equipment above its carrying value, and is not part of core operating activity.

#164 · Financial Statement Analysis

What is the formula for gross profit margin?

Answer

Gross profit margin = $(\text{Revenue} - \text{Cost of goods sold}) \div \text{Revenue} \times 100\%$. It measures the percentage of revenue retained after covering direct production costs.

#165 · Financial Statement Analysis

What is the formula for net profit margin?

Answer

Net profit margin = Net income ÷ Revenue × 100%.
It shows the percentage of each revenue dollar that flows through to the bottom line after all expenses, interest, and taxes.

#166 · Financial Statement Analysis

What is the general principle behind matching expense recognition to revenue?

Answer

The matching principle requires expenses directly tied to generating revenue, such as cost of goods sold, to be recognized in the same period as the related revenue, so that period income reflects the true economic effort behind those sales.

#167 · Financial Statement Analysis

How are period costs treated differently from product costs?

Answer

Product costs are capitalized as inventory and expensed as cost of goods sold when the related goods are sold. Period costs, such as administrative salaries, are expensed in the period incurred regardless of when related revenue is earned.

#168 · Financial Statement Analysis

What is the formula for basic earnings per share?

Answer

Basic EPS = $(\text{Net income} - \text{Preferred dividends}) \div \text{Weighted average number of common shares outstanding}$.

#169 · Financial Statement Analysis

What is the formula for diluted earnings per share using the if-converted method for convertible debt?

Answer

Diluted EPS = (Net income – Preferred dividends + Convertible debt interest × (1 – Tax rate)) ÷ (Weighted average shares + Shares from conversion of convertible debt), including the conversion only if it is dilutive.

#170 · Financial Statement Analysis

What is the treasury stock method used for in diluted EPS?

Answer

The treasury stock method estimates the net new shares added when options or warrants are exercised, assuming the proceeds from exercise are used to repurchase shares at the average market price, which reduces the dilutive share impact compared to a simple addition of all option shares.

#171 · Financial Statement Analysis

When is a convertible security excluded from the diluted EPS calculation?

Answer

A convertible security is excluded, and treated as antidilutive, when including it would increase diluted EPS or decrease a diluted loss per share, since only dilutive securities that reduce EPS are included in the diluted calculation.

#172 · Financial Statement Analysis

What distinguishes a simple capital structure from a complex capital structure?

Answer

A simple capital structure contains no potentially dilutive securities, such as convertible bonds, convertible preferred stock, or options, so only basic EPS is reported. A complex capital structure contains such securities and requires both basic and diluted EPS to be disclosed.

#173 · Financial Statement Analysis

What is a common-size income statement and how is it constructed?

Answer

Every line item on the income statement is expressed as a percentage of total revenue, allowing an analyst to compare cost structure, profitability, and expense trends across companies of different sizes or across time periods for the same company.

#174 · Financial Statement Analysis

What is a common-size balance sheet and how is it constructed?

Answer

A common-size balance sheet expresses every asset, liability, and equity line item as a percentage of total assets, allowing comparison of capital structure and asset composition across companies regardless of scale.

#175 · Financial Statement Analysis

What is the accounting equation that underlies the balance sheet?

Answer

Assets = Liabilities + Shareholders' equity. Every recorded transaction must keep this equation in balance, which is the foundation of double-entry accounting.

#176 · Financial Statement Analysis

How are current assets ordered on a classified balance sheet and what defines them?

Answer

Current assets are those expected to be converted to cash, sold, or consumed within one year or one operating cycle, whichever is longer, typically listed in order of liquidity: cash, marketable securities, receivables, inventory, and prepaid expenses.

#177 · Financial Statement Analysis

What is the formula for working capital?

Answer

Working capital = Current assets – Current liabilities. It measures a company's short-term liquidity buffer, the resources available to meet near-term obligations.

#178 · Financial Statement Analysis

What is the difference between an operating expense and a nonoperating expense on the income statement?

Answer

Operating expenses, such as selling, general and administrative costs, arise from the company's core business activities. Nonoperating expenses, such as interest expense or losses on asset sales, arise from financing or peripheral activities unrelated to core operations.

#179 · Financial Statement Analysis

What is comprehensive income and how does it differ from net income?

Answer

Comprehensive income includes net income plus other comprehensive income items, such as unrealized gains and losses on available-for-sale securities, foreign currency translation adjustments, and certain pension adjustments, that bypass the income statement but affect equity.

#180 · Financial Statement Analysis

What is the formula for return on equity?

Answer

Return on equity = $\text{Net income} \div \text{Average shareholders' equity} \times 100\%$. It measures the profit generated per dollar of equity capital invested by shareholders.

#181 · Financial Statement Analysis

What is the formula for return on assets?

Answer

Return on assets = $\text{Net income} \div \text{Average total assets} \times 100\%$. It measures how efficiently a company uses its total asset base to generate profit.

#182 · Financial Statement Analysis

How does an unusual or infrequent item differ from an item classified as discontinued operations?

Answer

An unusual or infrequent item, such as a restructuring charge, is reported within income from continuing operations before tax. Discontinued operations represent a component the company has disposed of or classified as held for sale, reported separately net of tax below income from continuing operations.

#183 · Financial Statement Analysis

What is the weighted average number of shares outstanding used for in EPS, and how is a stock split handled?

Answer

It is the time-weighted average of common shares outstanding during the period, adjusted for share issuances and repurchases based on the fraction of the period they were outstanding. A stock split or stock dividend is applied retroactively to all periods presented as if it had occurred at the start of the earliest period shown.

#184 · Financial Statement Analysis

What are the two general approaches to classifying expenses on the income statement?

Answer

The nature-of-expense method classifies costs by type, such as depreciation, raw materials, and employee benefits. The function-of-expense (cost of sales) method classifies costs by their business function, such as cost of goods sold and administrative expense, and requires additional nature-based disclosure in notes under IFRS.

#185 · Financial Statement Analysis

What is a contra asset account and give an example on the balance sheet?

Answer

A contra asset account carries a credit balance and reduces the reported value of its related asset account. Accumulated depreciation is the classic example, reducing gross property, plant and equipment to its net book value.

#186 · Financial Statement Analysis

What is the formula for the current ratio and what does it measure?

Answer

Current ratio = Current assets ÷ Current liabilities. It measures a company's ability to cover short-term obligations using assets expected to convert to cash within one year.

#187 · Financial Statement Analysis

How does a bill-and-hold arrangement affect revenue recognition timing?

Answer

Revenue on a bill-and-hold arrangement can only be recognized once control has effectively transferred to the buyer, requiring a substantive reason for the arrangement, separate identification of the goods as belonging to the buyer, and the goods being ready for immediate physical transfer, even though physical delivery is delayed.

#188 · Financial Statement Analysis

What is the difference between a multi-step and a single-step income statement format?

Answer

A multi-step income statement separates operating items from nonoperating items, presenting subtotals such as gross profit and operating income before reaching net income. A single-step income statement groups all revenues together and all expenses together, arriving at net income in one subtraction.

#189 · Financial Statement Analysis

What is the formula for the quick ratio (acid-test ratio)?

Answer

Quick ratio = (Cash + Marketable securities + Receivables) ÷ Current liabilities. It is a stricter liquidity measure than the current ratio because it excludes inventory and prepaid expenses, which are less readily convertible to cash.

#190 · Financial Statement Analysis

How should a long-term construction contract recognize revenue over time under the input or output method?

Answer

Revenue is recognized over time in proportion to progress toward completion, measured either by an input method, such as costs incurred to date relative to total estimated costs, or an output method, such as units delivered or milestones achieved, provided the contract meets the over-time recognition criteria.

#191 · Financial Statement Analysis

What is the effect on EPS of a company repurchasing its own shares (treasury stock) during the year?

Answer

A share repurchase reduces the weighted average number of shares outstanding for the portion of the year after the buyback, which increases EPS for a given level of net income, all else equal.

#192 · Financial Statement Analysis

What is the difference between accrued revenue and unearned (deferred) revenue?

Answer

Accrued revenue is recognized when a service has been performed or goods delivered before cash is received, creating a receivable. Unearned revenue arises when cash is received before the related good or service is delivered, creating a liability until the performance obligation is satisfied.

#193 · Financial Statement Analysis

What is total shareholders' equity composed of on the balance sheet?

Answer

Shareholders' equity typically includes common stock at par value, additional paid-in capital, retained earnings, treasury stock (a contra equity account), and accumulated other comprehensive income, summing to the residual claim on assets after liabilities.

#194 · Financial Statement Analysis

Current ratio formula

Answer

Current assets ÷ Current liabilities. Measures a firm's ability to meet short-term obligations with short-term assets; higher generally means more liquid.

#195 · Financial Statement Analysis

Quick ratio (acid-test) formula

Answer

$(\text{Cash} + \text{Marketable securities} + \text{Receivables}) \div \text{Current liabilities}$. Excludes inventory and prepaid expenses since they are less readily converted to cash.

#196 · Financial Statement Analysis

Cash ratio formula

Answer

$(\text{Cash} + \text{Cash equivalents} + \text{Marketable securities}) \div \text{Current liabilities}$. The most conservative liquidity measure, using only the most liquid assets.

#197 · Financial Statement Analysis

Defensive interval ratio formula

Answer

$(\text{Cash} + \text{Marketable securities} + \text{Receivables}) \div$
Daily cash expenditures. Estimates how many days
a company can operate using only its existing liquid
assets.

#198 · Financial Statement Analysis

Cash conversion cycle formula

Answer

Days of inventory on hand + Days of sales
outstanding minus Days of payables outstanding.
Measures the time between paying for inventory
and collecting cash from its sale.

#199 · Financial Statement Analysis

Debt-to-equity ratio formula

Answer

Total debt ÷ Total shareholders' equity. A solvency ratio showing how much debt financing is used relative to equity financing.

#200 · Financial Statement Analysis

Debt-to-assets ratio formula

Answer

Total debt ÷ Total assets. Shows the proportion of a company's assets financed with debt; higher values indicate greater financial leverage.

#201 · Financial Statement Analysis

Interest coverage ratio formula

Answer

$\text{EBIT} \div \text{Interest expense}$. Measures how many times a company can cover its interest payments from operating earnings; lower values signal higher default risk.

#202 · Financial Statement Analysis

Financial leverage ratio formula

Answer

$\text{Average total assets} \div \text{Average total equity}$. Also called the equity multiplier; it is the third component of the DuPont decomposition of ROE.

#203 · Financial Statement Analysis

Gross profit margin formula

Answer

Gross profit ÷ Revenue, expressed as a percentage. Shows the percentage of revenue retained after subtracting cost of goods sold.

#204 · Financial Statement Analysis

Operating profit margin formula

Answer

Operating income ÷ Revenue, expressed as a percentage. Measures profitability from core operations before interest and taxes.

#205 · Financial Statement Analysis

Net profit margin formula

Answer

Net income ÷ Revenue, expressed as a percentage. Shows how much of each sales dollar remains as profit after all expenses, interest, and taxes.

#206 · Financial Statement Analysis

Return on assets (ROA) formula

Answer

Net income ÷ Average total assets, expressed as a percentage. Measures how efficiently a company uses its assets to generate profit.

#207 · Financial Statement Analysis

Return on equity (ROE) formula

Answer

Net income ÷ Average shareholders' equity, expressed as a percentage. Measures the profit generated per dollar of equity capital invested by shareholders.

#208 · Financial Statement Analysis

DuPont 3-factor ROE decomposition

Answer

ROE = Net profit margin × Asset turnover × Financial leverage, where Net profit margin = Net income ÷ Revenue, Asset turnover = Revenue ÷ Average total assets, and Financial leverage = Average total assets ÷ Average total equity.

#209 · Financial Statement Analysis

DuPont 5-factor ROE decomposition

Answer

$ROE = \text{Tax burden} \times \text{Interest burden} \times \text{Operating margin} \times \text{Asset turnover} \times \text{Financial leverage}$,
where $\text{Tax burden} = \text{Net income} \div \text{Pretax income}$
and $\text{Interest burden} = \text{Pretax income} \div \text{EBIT}$. This splits net profit margin into three drivers to isolate the effects of taxes, interest, and operations.

#210 · Financial Statement Analysis

Total asset turnover ratio formula

Answer

$\text{Revenue} \div \text{Average total assets}$. An activity ratio measuring how efficiently a company generates sales from its total asset base.

#211 · Financial Statement Analysis

Inventory turnover ratio formula

Answer

Cost of goods sold ÷ Average inventory. Measures how many times inventory is sold and replaced over a period; higher turnover generally indicates efficient inventory management.

#212 · Financial Statement Analysis

Days of inventory on hand (DOH) formula

Answer

$365 \div \text{Inventory turnover ratio}$. Estimates the average number of days a company holds inventory before selling it.

#213 · Financial Statement Analysis

Receivables turnover ratio formula

Answer

$\text{Revenue} \div \text{Average receivables}$. Measures how efficiently a company collects cash from customers who bought on credit.

#214 · Financial Statement Analysis

Days of sales outstanding (DSO) formula

Answer

$365 \div \text{Receivables turnover ratio}$. Estimates the average number of days it takes to collect cash from credit sales.

#215 · Financial Statement Analysis

Payables turnover ratio formula

Answer

$\text{Purchases} \div \text{Average payables}$. Measures how quickly a company pays its suppliers; a lower turnover means the company takes longer to pay.

#216 · Financial Statement Analysis

Days of payables outstanding (DPO) formula

Answer

$365 \div \text{Payables turnover ratio}$. Estimates the average number of days a company takes to pay its suppliers.

#217 · Financial Statement Analysis

FIFO vs LIFO in a period of rising prices

Answer

Under FIFO, older lower-cost units are expensed first, so cost of goods sold is lower and ending inventory and net income are higher. Under LIFO, newer higher-cost units are expensed first, so cost of goods sold is higher and ending inventory and net income are lower.

#218 · Financial Statement Analysis

LIFO reserve definition

Answer

The difference between inventory reported under FIFO and inventory reported under LIFO. It is disclosed in footnotes and used to convert a LIFO firm's financials to a FIFO basis for comparison.

#219 · Financial Statement Analysis

Converting LIFO inventory to FIFO inventory

Answer

FIFO inventory equals LIFO inventory plus the LIFO reserve. This adjustment allows comparison of firms using different inventory cost flow assumptions.

#220 · Financial Statement Analysis

LIFO liquidation

Answer

Occurs when a LIFO firm sells more units than it purchases in a period, forcing it to dip into older, lower-cost layers of inventory. This produces an artificially lower cost of goods sold and inflated net income relative to normal operations.

#221 · Financial Statement Analysis

Straight-line depreciation formula

Answer

(Cost minus Residual value) ÷ Useful life. Allocates an equal amount of depreciation expense to each period of the asset's useful life.

#222 · Financial Statement Analysis

Double-declining balance depreciation formula

Answer

Depreciation rate = $2 \div \text{Useful life}$, multiplied by the asset's beginning-of-period net book value each year, ignoring residual value until book value approaches it. It is an accelerated method producing higher depreciation expense in early years.

#223 · Financial Statement Analysis

Direct method of preparing the cash flow statement

Answer

Reports actual cash inflows and outflows from operating activities, such as cash received from customers and cash paid to suppliers, rather than starting from net income. CFA Institute encourages this method for its clarity, though it is used less often in practice.

#224 · Financial Statement Analysis

Indirect method of preparing the cash flow statement

Answer

Starts with net income and adjusts for noncash items such as depreciation and amortization, plus changes in working capital accounts, to arrive at cash flow from operating activities. It is the method most companies actually use.

#225 · Financial Statement Analysis

Effective tax rate formula

Answer

Income tax expense ÷ Pretax income, expressed as a percentage. It reflects the actual average tax rate a company pays and can differ from the statutory tax rate due to permanent and temporary differences.

#226 · Corporate Issuers

What is corporate governance?

Answer

The system of internal controls, processes, and procedures by which a firm is directed and controlled, defining the relationships among shareholders, the board of directors, management, and other stakeholders.

#227 · Corporate Issuers

Define an agency relationship in the corporate context.

Answer

An arrangement in which a principal (owner) engages an agent (manager) to perform a service on the principal's behalf, delegating decision-making authority to that agent.

#228 · Corporate Issuers

What is an agency conflict (agency problem)?

Answer

A conflict of interest arising because agents (managers) may act in their own self-interest rather than in the best interest of the principals (shareholders) who hired them.

#229 · Corporate Issuers

What are agency costs?

Answer

The costs borne by shareholders that arise from conflicts of interest between shareholders and managers, including monitoring costs, bonding costs, and residual loss.

#230 · Corporate Issuers

What is the shareholder theory of corporate governance?

Answer

The view that a corporation's primary purpose is to maximize shareholder wealth and returns, with management acting strictly as agents for shareholders.

#231 · Corporate Issuers

What is the stakeholder theory of corporate governance?

Answer

The view that a corporation should balance the interests of all stakeholders, including shareholders, employees, customers, suppliers, and the community, not just shareholders alone.

#232 · Corporate Issuers

What is a classified (staggered) board?

Answer

A board structure in which directors are divided into groups elected in different years, so only a fraction of the board stands for election annually, which can entrench management and weaken shareholder control.

#233 · Corporate Issuers

What does board independence mean?

Answer

Board independence refers to directors having no material financial, family, or employment relationship with the company beyond their director role, allowing them to provide objective oversight of management.

#234 · Corporate Issuers

What is a dual-class share structure?

Answer

A share structure in which a company issues two or more classes of common stock with different voting rights, often letting founders retain control while owning a smaller percentage of cash flow rights.

#235 · Corporate Issuers

What is the difference between statutory voting and cumulative voting?

Answer

Under statutory voting each shareholder gets one vote per share for each director position; under cumulative voting shareholders can allocate all their votes to one or a few candidates, which helps minority shareholders elect a director.

#236 · Corporate Issuers

What is a poison pill takeover defense?

Answer

A shareholder rights plan that allows existing shareholders (excluding the acquirer) to purchase additional shares at a discount if a hostile bidder acquires a threshold stake, diluting the acquirer and making the takeover more costly.

#237 · Corporate Issuers

What is greenmail?

Answer

A defensive tactic in which a target company repurchases a hostile acquirer's shares at a premium price in exchange for the acquirer agreeing to end its takeover attempt.

#238 · Corporate Issuers

State the Modigliani-Miller Proposition I without taxes.

Answer

In perfect capital markets with no taxes, a firm's value is independent of its capital structure; the way a firm finances its operations does not affect its total value.

#239 · Corporate Issuers

State the Modigliani-Miller Proposition II without taxes.

Answer

The cost of equity increases linearly with the firm's debt-to-equity ratio, because financial leverage increases the risk borne by equity holders, exactly offsetting the benefit of using cheaper debt.

#240 · Corporate Issuers

How does MM Proposition I change when corporate taxes are introduced?

Answer

With corporate taxes, firm value increases with leverage because interest payments are tax deductible, creating a valuable interest tax shield:
Value of levered firm = Value of unlevered firm + (Tax rate $\times$ Debt).

#241 · Corporate Issuers

What is the static trade-off theory of capital structure?

Answer

The theory that a firm's optimal capital structure balances the tax benefits of debt against the increasing costs of financial distress and bankruptcy as leverage rises.

#242 · Corporate Issuers

What is pecking order theory?

Answer

A theory stating that firms prefer internal financing first, then debt, and equity issuance as a last resort, due to information asymmetry making external financing costlier, especially equity.

#243 · Corporate Issuers

What is the weighted average cost of capital (WACC) formula?

Answer

$WACC = (E \div V) \times r_e + (D \div V) \times r_d \times (1 - t)$, where E is market value of equity, D is market value of debt, $V = E + D$, r_e is cost of equity, r_d is cost of debt, and t is the marginal tax rate.

#244 · Corporate Issuers

How is the cost of debt typically estimated?

Answer

The cost of debt is usually estimated as the yield to maturity on the firm's existing long-term debt, then adjusted after tax as $r_d \times (1 - t)$ since interest expense is tax deductible.

#245 · Corporate Issuers

State the Capital Asset Pricing Model (CAPM) formula for cost of equity.

Answer

$r_e = R_f + \beta \times (R_m - R_f)$, where R_f is the risk-free rate, β is the stock's beta, and $(R_m - R_f)$ is the equity market risk premium.

#246 · Corporate Issuers

What is the dividend discount model approach to estimating cost of equity?

Answer

$r_e = (D_1 \div P_0) + g$, where D_1 is next year's expected dividend, P_0 is the current stock price, and g is the constant expected dividend growth rate.

#247 · Corporate Issuers

What is the marginal cost of capital?

Answer

The cost of raising one additional dollar of new capital, which typically rises as a company raises progressively larger amounts of financing because cheaper sources of capital are exhausted first.

#248 · Corporate Issuers

What is a breakpoint in the marginal cost of capital schedule?

Answer

The level of total new financing at which the cost of one or more capital components changes, causing the firm's marginal cost of capital to increase; calculated as $(\text{Amount of capital at which a source's cost changes}) \div (\text{Target weight of that source})$.

#249 · Corporate Issuers

What is the NPV (Net Present Value) decision rule for capital budgeting?

Answer

Accept a project if $NPV > 0$, since it is expected to increase firm value; reject if $NPV < 0$. NPV = sum of the present values of all expected after-tax cash flows minus the initial investment.

#250 · Corporate Issuers

What is the IRR (Internal Rate of Return) decision rule?

Answer

Accept a project if its IRR exceeds the required rate of return (cost of capital); the IRR is the discount rate at which NPV equals zero.

#251 · Corporate Issuers

When can the NPV and IRR methods give conflicting rankings for mutually exclusive projects?

Answer

Conflicts arise when projects differ substantially in size, timing of cash flows, or have unconventional cash flow patterns; in such cases NPV should be preferred because it directly measures value added to the firm.

#252 · Corporate Issuers

What is the payback period method?

Answer

The number of years required to recover the initial investment from a project's expected cash flows; it ignores the time value of money and cash flows occurring after the payback point.

#253 · Corporate Issuers

What is the discounted payback period?

Answer

The number of years required to recover the initial investment using discounted cash flows rather than nominal cash flows, addressing the time value of money weakness of the regular payback method but still ignoring cash flows after payback.

#254 · Corporate Issuers

What is the profitability index (PI)?

Answer

$PI = (\text{Present value of future cash flows} \div \text{Initial investment}) = 1 + (\text{NPV} \div \text{Initial investment})$; a project is acceptable when PI is greater than 1.

#255 · Corporate Issuers

What is working capital management?

Answer

The management of a firm's short-term assets and liabilities, primarily cash, receivables, inventory, and payables, to ensure the firm can meet its short-term obligations while operating efficiently.

#256 · Corporate Issuers

What is the cash conversion cycle (CCC) formula?

Answer

$$\text{CCC} = \text{Days of inventory on hand} + \text{Days of sales outstanding} - \text{Days of payables outstanding}$$

A shorter CCC means the firm converts investments in inventory back into cash more quickly.

#257 · Corporate Issuers

What is the current ratio?

Answer

Current ratio = Current assets ÷ Current liabilities; it measures a firm's ability to meet short-term obligations using assets expected to be converted to cash within one year.

#258 · Corporate Issuers

What is the quick ratio (acid-test ratio)?

Answer

Quick ratio = (Cash + Short-term marketable investments + Receivables) ÷ Current liabilities; it excludes inventory to give a more conservative measure of short-term liquidity.

#259 · Corporate Issuers

What are the primary sources of liquidity for a firm?

Answer

Primary sources include cash balances, short-term funds from operating cash flow, effective management of receivables and inventory, and existing bank credit lines available on short notice.

#260 · Corporate Issuers

What is the difference between a company's business risk and its financial risk?

Answer

Business risk is the uncertainty in operating earnings arising from a firm's operations and industry; financial risk is the additional risk borne by equity holders due to the firm's use of debt financing, which increases the volatility of earnings available to shareholders.

#261 · Equity Investments

What are the two main types of equity securities?

Answer

Common shares and preferred shares. Common shares carry voting rights and a residual claim on earnings and assets; preferred shares generally have no voting rights but receive a fixed dividend with priority over common shareholders.

#262 · Equity Investments

What rights does a common shareholder typically have?

Answer

The right to vote on corporate matters such as electing directors and major transactions, the right to share in declared dividends, and a residual claim on assets in liquidation after creditors and preferred shareholders are paid.

#263 · Equity Investments

Define cumulative preferred stock.

Answer

Preferred stock on which any unpaid (skipped) dividends accumulate as arrears and must be paid in full to preferred shareholders before any dividend can be paid to common shareholders.

#264 · Equity Investments

Define callable preferred stock.

Answer

Preferred stock that gives the issuer the right, but not the obligation, to redeem the shares from investors at a specified call price after a stated date, typically used when the issuer expects rates or its cost of capital to fall.

#265 · Equity Investments

Define putable preferred stock.

Answer

Preferred stock that gives the holder the right to sell the shares back to the issuer at a specified price, which reduces the investor's downside risk and price volatility compared with nonputable preferred.

#266 · Equity Investments

What is convertible preferred stock?

Answer

Preferred stock that can be exchanged for a fixed number of common shares of the issuing company at the holder's option, giving upside participation in the common stock while retaining a preferred dividend claim.

#267 · Equity Investments

What distinguishes participating preferred stock?

Answer

In addition to its fixed dividend, participating preferred stock entitles holders to an extra dividend if the company's profits exceed a specified level, and often to a share of liquidation proceeds above par value.

#268 · Equity Investments

What is the difference between public equity and private equity?

Answer

Public equity trades on organized exchanges or over-the-counter markets and is subject to public disclosure requirements, while private equity is issued through private placements to a limited number of investors and is not registered for public trading.

#269 · Equity Investments

What is venture capital?

Answer

Private equity capital provided to early-stage, high-growth-potential companies in exchange for an ownership stake, typically funding product development or expansion before the company is ready for public markets.

#270 · Equity Investments

What are the two broad categories of equity market structures?

Answer

Quote-driven (dealer) markets, where investors trade with dealers who post bid and ask prices, and order-driven markets, where buy and sell orders from investors are matched directly, often through an order book.

#271 · Equity Investments

What is a primary market versus a secondary market?

Answer

The primary market is where new securities are issued and sold to investors for the first time, such as in an IPO, while the secondary market is where previously issued securities are traded among investors.

#272 · Equity Investments

What is a price-weighted equity index?

Answer

An index in which each constituent's weight is determined by its share price divided by the sum of all prices, so higher-priced stocks have a greater influence on the index regardless of company size. The Dow Jones Industrial Average is a classic example.

#273 · Equity Investments

What is a market-capitalization-weighted equity index?

Answer

An index in which each constituent's weight is based on its total market value, calculated as share price times shares outstanding, so larger companies have a greater influence on index performance.

#274 · Equity Investments

What is a float-adjusted market-capitalization-weighted index?

Answer

A market-cap-weighted index that uses only the shares available for public trading, excluding closely held or restricted shares, when calculating each constituent's weight and market value.

#275 · Equity Investments

What is an equal-weighted equity index?

Answer

An index that assigns the same weight to each constituent regardless of price or market capitalization, which requires periodic rebalancing to restore equal weights as prices change.

#276 · Equity Investments

What is a fundamental-weighted equity index?

Answer

An index that weights constituents by measures of company size or economic footprint other than market price, such as revenue, earnings, book value, or dividends, rather than by market capitalization.

#277 · Equity Investments

What is index reconstitution?

Answer

The periodic process of changing the constituent securities that make up an index to reflect additions, deletions, and mergers, ensuring the index continues to represent its intended target market.

#278 · Equity Investments

What is index rebalancing?

Answer

The periodic adjustment of the weights of existing index constituents back to their target levels as defined by the index's weighting methodology, without necessarily changing which securities are included.

#279 · Equity Investments

What does the weak form of the efficient market hypothesis (EMH) state?

Answer

Current stock prices fully reflect all past price and trading volume information, so technical analysis based on historical price patterns cannot be used to consistently earn abnormal returns.

#280 · Equity Investments

What does the semi-strong form of the EMH state?

Answer

Current stock prices fully reflect all publicly available information, including past prices and fundamental data such as financial statements, so neither technical nor fundamental analysis can consistently generate abnormal returns.

#281 · Equity Investments

What does the strong form of the EMH state?

Answer

Current stock prices fully reflect all public and private (insider) information, meaning even investors with material nonpublic information could not consistently earn abnormal returns.

#282 · Equity Investments

What is the momentum anomaly in market efficiency?

Answer

A pattern in which securities that have performed well over a recent period continue to outperform, and those that have performed poorly continue to underperform, contradicting weak-form market efficiency.

#283 · Equity Investments

What is overreaction bias as a market anomaly?

Answer

A behavioral pattern in which investors overreact to new information, pushing prices too far in the direction of the news before they later correct, creating a predictable reversal that challenges market efficiency.

#284 · Equity Investments

What is the size effect anomaly?

Answer

The empirical observation that small-capitalization stocks have historically earned higher risk-adjusted returns than large-capitalization stocks, which is difficult to reconcile with efficient markets.

#285 · Equity Investments

List Porter's Five Forces framework for industry analysis.

Answer

Threat of new entrants, bargaining power of suppliers, bargaining power of buyers, threat of substitute products or services, and rivalry among existing competitors.

#286 · Equity Investments

According to Porter's framework, what raises the threat of new entrants?

Answer

Low barriers to entry, such as minimal capital requirements, easy access to distribution channels, low switching costs for customers, and limited economies of scale enjoyed by incumbents.

#287 · Equity Investments

What increases buyer bargaining power in Porter's framework?

Answer

Buyer power rises when buyers are concentrated or purchase in large volumes, the product is undifferentiated, switching costs are low, and buyers have full information or the ability to backward integrate.

#288 · Equity Investments

What increases supplier bargaining power in Porter's framework?

Answer

Supplier power rises when suppliers are few or concentrated, their input is critical and lacks substitutes, switching suppliers is costly, and suppliers could plausibly forward integrate into the buyer's industry.

#289 · Equity Investments

What are the typical stages of the industry life cycle?

Answer

Embryonic (introduction), growth, shakeout, mature, and decline, each characterized by different patterns of sales growth, profitability, competitive intensity, and barriers to entry.

#290 · Equity Investments

What characterizes the growth stage of the industry life cycle?

Answer

Rapid increases in unit sales and revenue as products gain widespread acceptance, improving but still uneven profitability across firms, and relatively low competitive rivalry since demand growth exceeds industry capacity.

#291 · Equity Investments

What characterizes the shakeout stage of the industry life cycle?

Answer

Growth slows and becomes more dependent on overall economic growth, competition intensifies for market share, profit margins narrow, and weaker competitors exit or are acquired.

#292 · Equity Investments

What characterizes the mature stage of the industry life cycle?

Answer

Industry growth roughly matches overall economic growth, market shares stabilize among a few dominant firms, barriers to entry are high, and profitability is stable but no longer rapidly expanding.

#293 · Equity Investments

What is a barrier to entry, and why does it matter for industry analysis?

Answer

A barrier to entry is a factor, such as high capital requirements, brand loyalty, or regulation, that makes it difficult for new competitors to enter an industry; high barriers protect incumbents' profitability and market share.

#294 · Equity Investments

What is the Gordon growth model formula?

Answer

$V_0 = D_1 \div (r - g)$, where D_1 is next year's expected dividend, r is the required rate of return, and g is the constant expected dividend growth rate. It values a stock as the present value of a perpetually growing dividend stream.

#295 · Equity Investments

What condition must hold for the Gordon growth model to produce a meaningful (positive, finite) value?

Answer

The required return must exceed the growth rate ($r > g$). If $g \geq r$, the model produces a negative or undefined value and cannot be used.

#296 · Equity Investments

How do you find D1 if only the current dividend D0 is given?

Answer

$D1 = D0 \times (1 + g)$, where g is the constant expected growth rate. D1 is always the dividend expected to be paid one period from today.

#297 · Equity Investments

What is the implied growth rate formula when solving the Gordon growth model for g?

Answer

$g = r - (D1 \div V0)$. Rearranging the Gordon growth model isolates g given the required return and the current price or value.

#298 · Equity Investments

What is the implied required return formula when solving the Gordon growth model for r?

Answer

$r = (D1 \div V0) + g$. This is the sum of the dividend yield ($D1 \div V0$) and the expected growth rate, a common way to estimate the cost of equity.

#299 · Equity Investments

What is the dividend displacement of earnings problem in the Gordon growth model?

Answer

A firm that retains more earnings to fund higher growth pays a lower current dividend, so higher assumed g does not always raise value; the model captures a trade-off between current dividend size and future growth.

#300 · Equity Investments

What are the two components of total expected return implied by the Gordon growth model?

Answer

Expected total return equals the dividend yield ($D_1 \div V_0$) plus the expected capital gains yield, which under constant growth equals g .

#301 · Equity Investments

What is the sustainable growth rate formula?

Answer

$g = b \times \text{ROE}$, where b is the earnings retention rate (1 minus the dividend payout ratio) and ROE is return on equity. It estimates the growth rate a firm can sustain without external equity financing.

#302 · Equity Investments

How is the dividend payout ratio defined?

Answer

Payout ratio = dividends per share $\div$ earnings per share. It is the complement of the retention rate b , so $b = 1 - \text{payout ratio}$.

#303 · Equity Investments

What is the two-stage dividend discount model used for?

Answer

It values a stock with an initial high-growth period followed by a stable, sustainable long-term growth rate; value equals the present value of the high-growth stage dividends plus the present value of a Gordon growth terminal value at the end of that stage.

#304 · Equity Investments

How is the terminal value calculated at the end of a multistage dividend discount model?

Answer

Terminal value = $D_{n+1} \div (r - g_{\text{stable}})$, applying the Gordon growth formula to the first stable-growth-stage dividend, then discounting that terminal value back to the present at r .

#305 · Equity Investments

What is the justified leading P/E ratio formula under the Gordon growth model?

Answer

Leading P/E = $(D1 \div E1) \div (r - g)$, which equals the expected dividend payout ratio divided by the required return minus the growth rate.

#306 · Equity Investments

What is the justified trailing P/E ratio formula under the Gordon growth model?

Answer

Trailing P/E = $[(1 - b) \times (1 + g)] \div (r - g)$, where $(1 - b)$ is the current payout ratio, b is the retention rate, and g is the constant growth rate.

#307 · Equity Investments

What is the basic price to earnings (P/E) ratio formula?

Answer

P/E = price per share ÷ earnings per share. It measures how much investors pay per dollar of a company's earnings and is the most widely used equity valuation multiple.

#308 · Equity Investments

What is the difference between trailing P/E and leading P/E?

Answer

Trailing P/E uses earnings per share from the most recent four quarters (or fiscal year); leading (forward) P/E uses expected earnings per share for the next four quarters or next fiscal year.

#309 · Equity Investments

What is normalized earnings per share and why is it used?

Answer

Normalized EPS is the level of EPS a company would achieve under mid-cycle economic conditions, removing the effects of business-cycle peaks and troughs, used to compute a more meaningful P/E for cyclical companies.

#310 · Equity Investments

What is the PEG ratio?

Answer

$$\text{PEG} = \text{P/E} \div \text{expected earnings growth rate}$$

(expressed as a whole number, not a decimal). A lower PEG suggests a stock is more attractively priced relative to its growth prospects, though it assumes a linear P/E-to-growth relationship.

#311 · Equity Investments

What is the price to book (P/B) ratio formula?

Answer

$P/B = \text{market price per share} \div \text{book value per share}$, where book value per share is total shareholders' equity (excluding preferred equity) divided by shares outstanding.

#312 · Equity Investments

What are common limitations of the price to book ratio?

Answer

Book value can be distorted by different accounting methods, does not reflect the value of intangible assets like brand or human capital, and can become negative or meaningless for firms with substantial share buybacks or losses.

#313 · Equity Investments

What is the price to sales (P/S) ratio formula?

Answer

$P/S = \text{market price per share} \div \text{sales per share}$ (or $\text{market capitalization} \div \text{total revenue}$). It is useful for valuing companies with negative earnings since revenue is rarely negative.

#314 · Equity Investments

What is a key advantage of the price to sales ratio over the P/E ratio?

Answer

Sales figures are less subject to accounting distortions and manipulation than earnings, and P/S remains meaningful even when a company has negative or highly volatile earnings.

#315 · Equity Investments

What is the price to cash flow (P/CF) ratio?

Answer

$P/CF = \text{market price per share} \div \text{cash flow per share}$. It avoids some of the accounting distortions in earnings and is harder for management to manipulate than net income.

#316 · Equity Investments

What is free cash flow to the firm (FCFF)?

Answer

FCFF is the cash flow available to all providers of capital (both debt and equity holders) after the firm pays operating expenses, taxes, and makes necessary investments in working capital and fixed capital.

#317 · Equity Investments

What is the formula for FCFF starting from net income?

Answer

$$\text{FCFF} = \text{NI} + \text{NCC} + \text{Int} \times (1 - \text{tax rate}) - \text{FCInv} - \text{WCInv}$$
, where NCC is non-cash charges, Int is interest expense, FCInv is fixed capital investment, and WCInv is working capital investment.

#318 · Equity Investments

What is free cash flow to equity (FCFE)?

Answer

FCFE is the cash flow available to common shareholders only, after operating expenses, interest, taxes, net borrowing, and investments in working capital and fixed capital have been paid.

#319 · Equity Investments

What is the formula for FCFE starting from FCFF?

Answer

$$\text{FCFE} = \text{FCFF} - \text{Int} \times (1 - \text{tax rate}) + \text{net borrowing},$$
where net borrowing is new debt issued minus debt repaid.

#320 · Equity Investments

How do you value the firm using FCFF, and how do you get to equity value?

Answer

Firm value = present value of all future FCFF discounted at the weighted average cost of capital (WACC). Equity value equals firm value minus the market value of debt (and any other non-equity claims).

#321 · Equity Investments

How do you value equity directly using FCFE?

Answer

Equity value = present value of all future FCFE discounted at the required rate of return on equity (r), then equity value per share = total equity value $\div$ shares outstanding.

#322 · Equity Investments

What discount rate is used for FCFF versus FCFE valuation?

Answer

FCFF is discounted at the weighted average cost of capital (WACC), since it represents cash flow to all capital providers. FCFE is discounted at the required return on equity, since it represents cash flow to shareholders only.

#323 · Equity Investments

When is a free-cash-flow-based valuation model preferred over the dividend discount model?

Answer

When a company pays no dividends or dividends that are unrelated to earnings capacity, or when an analyst takes a control perspective and wants to value the cash flow available for discretionary use rather than only what is actually distributed.

#324 · Equity Investments

What is the constant growth FCFE valuation model formula?

Answer

Equity value = $FCFE_1 \div (r - g)$, applying the same Gordon growth logic as the dividend discount model but to free cash flow to equity instead of dividends.

#325 · Equity Investments

Why might price multiples based on comparables give a different value estimate than a discounted cash flow model?

Answer

Multiples reflect current market pricing of similar companies, which can embed mispricing or differing risk and growth assumptions across the peer group, while a DCF model derives value directly from a company's own projected fundamentals and required return.

#326 · Fixed Income

What is a bond's par value?

Answer

The principal amount the issuer promises to repay the bondholder at maturity, also called face value or maturity value. Coupon payments are typically calculated as a percentage of par value.

#327 · Fixed Income

What is the coupon rate?

Answer

The annual interest rate stated on a bond, used to calculate the periodic coupon payment as coupon rate $\times$ par value. It is fixed at issuance for a plain vanilla fixed-rate bond.

#328 · Fixed Income

Define a zero-coupon bond.

Answer

A bond that pays no periodic coupons and is issued at a discount to par value, with the investor's entire return coming from the difference between the purchase price and par value at maturity.

#329 · Fixed Income

What is a floating-rate note (FRN)?

Answer

A bond whose coupon rate resets periodically based on a reference rate plus a fixed quoted margin, for example reference rate + 0.75%. Its price stays closer to par than a fixed-rate bond because the coupon adjusts with market rates.

#330 · Fixed Income

What is a bond indenture?

Answer

The legal contract between the bond issuer and bondholders that specifies the terms of the bond, including obligations, restrictive covenants, and the rights of both parties.

#331 · Fixed Income

Distinguish an affirmative covenant from a negative covenant.

Answer

An affirmative covenant requires the issuer to take certain actions, such as paying taxes and maintaining insurance. A negative covenant restricts the issuer from certain actions, such as limiting additional debt issuance or asset sales.

#332 · Fixed Income

What is a secured bond versus an unsecured bond?

Answer

A secured bond is backed by specific collateral pledged by the issuer, giving bondholders a claim on that asset if the issuer defaults. An unsecured bond, or debenture, is backed only by the issuer's general creditworthiness.

#333 · Fixed Income

What is a callable bond?

Answer

A bond that gives the issuer the right to redeem it before maturity at a specified call price, typically exercised when interest rates fall so the issuer can refinance at a lower rate. This benefits the issuer and works against the bondholder.

#334 · Fixed Income

What is a putable bond?

Answer

A bond that gives the bondholder the right to sell the bond back to the issuer at a specified price before maturity, typically exercised when interest rates rise. This benefits the bondholder and works against the issuer.

#335 · Fixed Income

What is a convertible bond?

Answer

A bond that gives the bondholder the right to exchange it for a specified number of the issuer's common shares. It typically carries a lower coupon rate than an otherwise identical non-convertible bond because of the conversion option's value.

#336 · Fixed Income

What are the primary and secondary bond markets?

Answer

The primary market is where newly issued bonds are sold directly to investors, often through underwriters. The secondary market is where previously issued bonds are traded among investors, providing liquidity and price discovery.

#337 · Fixed Income

Distinguish a public offering from a private placement.

Answer

A public offering registers bonds with regulators and sells them to the general investing public. A private placement sells bonds directly to a small group of qualified institutional investors without public registration, usually with less liquidity.

#338 · Fixed Income

What is an underwritten offering?

Answer

An offering in which an investment bank purchases the entire bond issue from the issuer at a set price and then resells it to investors, bearing the risk of not being able to resell at a profit.

#339 · Fixed Income

What is the current yield formula?

Answer

Current yield = annual coupon payment ÷ current bond price. It measures income return only and ignores capital gains or losses and the time value of money.

#340 · Fixed Income

What is yield to maturity (YTM)?

Answer

The internal rate of return that equates the present value of a bond's remaining cash flows to its current market price, assuming the bond is held to maturity and all coupons are reinvested at the YTM.

#341 · Fixed Income

What is yield to call (YTC)?

Answer

The internal rate of return on a callable bond assuming it is redeemed by the issuer at the first call date and call price rather than held to maturity, calculated the same way as YTM but using the call date and call price as the final cash flow.

#342 · Fixed Income

What is yield to worst?

Answer

The lowest possible yield among all the yield measures calculated for a callable bond's various call dates and yield to maturity. It represents the most conservative return an investor should assume.

#343 · Fixed Income

State the inverse price/yield relationship for bonds.

Answer

Bond prices and yields move in opposite directions. When market interest rates rise, bond prices fall, and when market interest rates fall, bond prices rise, because future cash flows are discounted at the new required yield.

#344 · Fixed Income

When does a bond trade at a premium, discount, or par?

Answer

A bond trades at a premium when its coupon rate exceeds the market yield, at a discount when its coupon rate is below the market yield, and at par when its coupon rate equals the market yield.

#345 · Fixed Income

What is the basic bond pricing formula?

Answer

Price = sum of each coupon payment $\div (1 + \text{yield})^t$ for each period t , plus par value $\div (1 + \text{yield})^n$, where n is the number of periods to maturity.

#346 · Fixed Income

What is a bond's full price (dirty price)?

Answer

The flat price of a bond plus accrued interest owed to the seller since the last coupon payment date. Buyers pay the full price at settlement even though the flat price is what is typically quoted.

#347 · Fixed Income

How is accrued interest calculated?

Answer

Accrued interest = periodic coupon payment × (days since last coupon payment ÷ days in the coupon period). It compensates the seller for interest earned but not yet paid at the time of sale.

#348 · Fixed Income

What is the flat price (clean price) of a bond?

Answer

The quoted price of a bond excluding accrued interest. Flat price = full price minus accrued interest, and it is the price typically reported by financial data sources.

#349 · Fixed Income

What is the yield curve?

Answer

A graphical depiction of the relationship between yields and maturities for bonds of similar credit quality, most commonly plotted for government benchmark bonds across the maturity spectrum.

#350 · Fixed Income

Describe a normal (upward-sloping) yield curve.

Answer

A yield curve where longer-maturity bonds have higher yields than shorter-maturity bonds, typically reflecting expectations of economic growth and a premium demanded for holding longer-term risk.

#351 · Fixed Income

Describe an inverted yield curve.

Answer

A yield curve where shorter-maturity bonds have higher yields than longer-maturity bonds. It is often viewed as a signal that investors expect slowing economic growth or falling future interest rates.

#352 · Fixed Income

What is the par yield curve?

Answer

A yield curve that plots the yields to maturity of hypothetical bonds priced at exactly par for each maturity, showing the coupon rate a newly issued bond at that maturity would need to trade at par.

#353 · Fixed Income

What is the spot rate (zero-coupon) yield curve?

Answer

A yield curve plotting the yields on zero-coupon bonds of different maturities. Each spot rate is the discount rate for a single future cash flow occurring at that maturity, used to value each cash flow of a coupon bond individually.

#354 · Fixed Income

What is a forward rate?

Answer

An interest rate for a future period that is implied by the current spot rate curve, representing the rate the market expects to prevail for borrowing or lending starting at a future date.

#355 · Fixed Income

What is the pure expectations theory of the term structure?

Answer

A theory stating that current long-term interest rates are determined solely by the market's expectations of future short-term interest rates, with forward rates being unbiased predictors of future spot rates.

#356 · Fixed Income

What is the liquidity preference theory of the term structure?

Answer

A theory stating that investors demand a liquidity premium for holding longer-maturity bonds to compensate for greater price risk and reduced liquidity, causing longer-term yields to exceed what pure rate expectations alone would suggest.

#357 · Fixed Income

What is a bond's yield spread?

Answer

The difference in yield between a bond and a benchmark bond of similar maturity, usually a government security, expressed in basis points and used to measure relative credit and liquidity risk.

#358 · Fixed Income

What is the constant-yield price trajectory?

Answer

The path a bond's price follows over time toward par value, assuming yields remain unchanged, as a premium bond's price gradually declines toward par and a discount bond's price gradually rises toward par as maturity approaches.

#359 · Fixed Income

What is Macaulay duration?

Answer

Macaulay duration is the weighted average time, in years, until a bond's cash flows are received, with weights based on the present value of each cash flow relative to the bond's total price.

#360 · Fixed Income

What is modified duration?

Answer

Modified duration measures the approximate percentage change in a bond's price for a 1% change in yield. It equals Macaulay duration divided by (1 plus the periodic yield).

#361 · Fixed Income

How do you approximate a bond's percentage price change using modified duration?

Answer

Percentage price change is approximately equal to negative modified duration multiplied by the change in yield. The negative sign reflects the inverse relationship between price and yield.

#362 · Fixed Income

What does convexity measure?

Answer

Convexity measures the curvature of the price-yield relationship, capturing how duration itself changes as yield changes. It improves the accuracy of price change estimates beyond what duration alone predicts.

#363 · Fixed Income

Why does duration alone underestimate a bond's price increase when yields fall?

Answer

Duration assumes a linear price-yield relationship, but the true relationship is convex. For a yield decrease, the actual price increase is larger than the duration-only estimate because of positive convexity.

#364 · Fixed Income

How does coupon rate affect duration?

Answer

A lower coupon rate increases duration because more of the bond's total value is concentrated in the final principal payment, lengthening the weighted average time to receive cash flows.

#365 · Fixed Income

How does maturity affect duration, holding other factors constant?

Answer

Longer maturity generally increases duration because cash flows are received further in the future, increasing the weighted average time to receipt and the bond's sensitivity to yield changes.

#366 · Fixed Income

How does yield to maturity affect duration?

Answer

A higher yield to maturity decreases duration because higher discount rates reduce the present value weight of distant cash flows relative to nearer ones, shortening the weighted average time.

#367 · Fixed Income

What is the duration of a zero-coupon bond?

Answer

The duration of a zero-coupon bond equals its maturity in years, since it has only one cash flow, the principal repayment at maturity.

#368 · Fixed Income

What is effective duration and when is it preferred over modified duration?

Answer

Effective duration measures price sensitivity to a parallel shift in the benchmark yield curve using actual price changes from a valuation model. It is preferred for bonds with embedded options because cash flows can change with yield.

#369 · Fixed Income

How does an embedded call option affect a bond's effective duration?

Answer

A callable bond's effective duration is lower than an otherwise identical option-free bond because the call option limits price appreciation when yields fall, as the issuer is likely to redeem the bond early.

#370 · Fixed Income

What is negative convexity and which bonds typically exhibit it?

Answer

Negative convexity occurs when price appreciation is limited as yields fall while price depreciation is not limited as yields rise. Callable bonds and mortgage-backed securities with prepayment risk typically exhibit it.

#371 · Fixed Income

What is credit risk?

Answer

Credit risk is the risk of loss from a borrower failing to make timely interest or principal payments as promised. It includes both default risk and the risk of a credit rating downgrade.

#372 · Fixed Income

What is default risk versus spread risk?

Answer

Default risk is the probability that an issuer fails to meet its debt obligations. Spread risk is the risk that a bond's price falls because its credit spread widens, even without an actual default.

#373 · Fixed Income

What are the four Cs used to assess credit risk according to the 'four Cs of credit analysis'?

Answer

The four Cs are capacity to pay, collateral backing the debt, covenants restricting borrower behavior, and character of management. Together they help assess an issuer's willingness and ability to repay debt.

#374 · Fixed Income

What distinguishes investment-grade bonds from high-yield (non-investment-grade) bonds?

Answer

Investment-grade bonds are rated Baa3/BBB- or higher by major rating agencies, reflecting lower default risk. High-yield bonds are rated below that threshold and carry higher default risk and higher yields to compensate.

#375 · Fixed Income

Name the three major credit rating agencies referenced in CFA curriculum.

Answer

The three major credit rating agencies are Moody's Investors Service, Standard & Poor's (S&P;), and Fitch Ratings.

#376 · Fixed Income

What is a credit spread?

Answer

A credit spread is the difference in yield between a corporate or other credit-risky bond and a benchmark government bond of similar maturity. It compensates investors for default risk, liquidity risk, and other risks.

#377 · Fixed Income

What typically happens to credit spreads during an economic downturn?

Answer

Credit spreads typically widen during an economic downturn as investors demand greater compensation for increased default risk and reduced market liquidity, causing risky bond prices to fall relative to government bonds.

#378 · Fixed Income

What is a credit rating downgrade's typical effect on bond price?

Answer

A credit rating downgrade typically causes a bond's price to fall because investors perceive higher default risk, requiring a higher yield and wider credit spread to compensate.

#379 · Fixed Income

What is seniority ranking in the context of credit risk?

Answer

Seniority ranking determines the priority order in which bondholders are repaid in the event of default or liquidation, with senior secured debt paid before senior unsecured, then subordinated debt.

#380 · Fixed Income

What is recovery rate?

Answer

Recovery rate is the percentage of a bond's par value that investors receive if the issuer defaults, reflecting proceeds recovered after liquidation or restructuring relative to the original claim.

#381 · Fixed Income

What is securitization?

Answer

Securitization is the process of pooling financial assets such as mortgages or loans and converting them into tradable securities backed by the cash flows from the underlying asset pool.

#382 · Fixed Income

What is a mortgage-backed security (MBS)?

Answer

A mortgage-backed security is a bond backed by a pool of residential or commercial mortgage loans, where investors receive cash flows derived from the underlying borrowers' principal and interest payments.

#383 · Fixed Income

What is an asset-backed security (ABS)?

Answer

An asset-backed security is a bond backed by a pool of financial assets other than mortgages, such as auto loans, credit card receivables, or student loans, with cash flows passed through to investors.

#384 · Fixed Income

What is prepayment risk in mortgage-backed securities?

Answer

Prepayment risk is the risk that homeowners repay mortgage principal earlier than scheduled, often by refinancing when rates fall, which returns investor principal sooner than expected and at a time when reinvestment rates are lower.

#385 · Fixed Income

What is a special purpose entity (SPE) in securitization?

Answer

A special purpose entity is a separate legal entity created to hold the pooled assets and issue securities backed by them, isolating those assets from the originator's other creditors and bankruptcy risk.

#386 · Fixed Income

What is tranching in a securitization structure?

Answer

Tranching divides a securitization into multiple classes of securities with different priority claims on cash flows and different risk levels, allowing senior tranches to be paid before subordinate tranches absorb losses first.

#387 · Fixed Income

Why does securitization generally benefit borrowers and the broader credit market?

Answer

Securitization increases the pool of funds available for lending by allowing originators to sell loans and free up capital, often lowering borrowing costs and increasing credit availability.

#388 · Fixed Income

How does the relationship between bond price and yield change as duration increases?

Answer

As duration increases, a bond's price becomes more sensitive to changes in yield, meaning a given change in yield produces a larger percentage change in price.

#389 · Fixed Income

Why do investors care about both duration and convexity when assessing interest rate risk?

Answer

Duration provides a first-order linear approximation of price sensitivity to yield changes, while convexity captures the curvature that duration misses, together giving a more accurate estimate especially for large yield changes.

#390 · Fixed Income

What is the general relationship between credit risk and required yield?

Answer

Higher perceived credit risk requires a higher yield to compensate investors for the greater probability of default and potential loss, resulting in a wider credit spread over a comparable risk-free bond.

#391 · Derivatives

What is a derivative?

Answer

A financial instrument whose value is derived from the value of an underlying asset, rate, or index. It creates a contractual relationship between two parties without requiring ownership of the underlying itself.

#392 · Derivatives

Define a forward contract.

Answer

A bilateral, customized over the counter agreement in which one party agrees to buy and the other to sell an underlying asset at a fixed price on a specified future date. It carries counterparty credit risk and is not marked to market daily.

#393 · Derivatives

How does a futures contract differ from a forward contract?

Answer

Futures are standardized, exchange traded contracts that are marked to market daily through a clearinghouse, which largely eliminates counterparty credit risk. Forwards are private, customized OTC agreements settled only at expiration.

#394 · Derivatives

What is the payoff at expiration for the long position in a forward contract?

Answer

The payoff to the long is the spot price of the underlying at expiration minus the forward price agreed at initiation. The short's payoff is the mirror image, the forward price minus the spot price.

#395 · Derivatives

What is a swap?

Answer

An OTC agreement between two parties to exchange a series of cash flows over time, typically based on different reference rates or asset returns applied to a notional principal that is not usually exchanged.

#396 · Derivatives

Describe a plain vanilla interest rate swap.

Answer

One party pays a fixed rate and receives a floating rate, such as a reference rate, on a notional principal, while the counterparty pays floating and receives fixed. Only net interest payments are exchanged, not principal.

#397 · Derivatives

What is a call option?

Answer

A contract giving the buyer the right, but not the obligation, to buy the underlying asset at a specified exercise price on or before expiration. The buyer pays a premium for this right.

#398 · Derivatives

What is a put option?

Answer

A contract giving the buyer the right, but not the obligation, to sell the underlying asset at a specified exercise price on or before expiration. The buyer pays a premium for this right.

#399 · Derivatives

What is the maximum loss for the buyer of a call option?

Answer

The maximum loss is limited to the premium paid for the option, which occurs if the option expires out of the money and is never exercised.

#400 · Derivatives

What is the maximum loss for the writer (seller) of an uncovered call option?

Answer

The loss is theoretically unlimited because the underlying asset price can rise without bound, and the writer must deliver the asset at the fixed exercise price regardless of how high the spot price rises.

#401 · Derivatives

When is a call option in the money?

Answer

A call option is in the money when the underlying asset's spot price is greater than the exercise price, meaning immediate exercise would produce a positive payoff.

#402 · Derivatives

When is a put option in the money?

Answer

A put option is in the money when the underlying asset's spot price is less than the exercise price, meaning immediate exercise would produce a positive payoff.

#403 · Derivatives

What does at the money mean for an option?

Answer

An option is at the money when the underlying asset's current spot price equals the exercise price, so immediate exercise would produce a payoff of zero.

#404 · Derivatives

Define intrinsic value of an option.

Answer

The value the option would have if exercised immediately, equal to the greater of zero or the difference between the spot price and exercise price in the option's favorable direction. It can never be negative.

#405 • Derivatives

Define time value (speculative value) of an option.

Answer

The portion of an option's premium above its intrinsic value, reflecting the possibility that the option will become more valuable before expiration due to future price movement and remaining time to expiry.

#406 • Derivatives

How does time value change as expiration approaches?

Answer

Time value generally decreases as expiration nears, a phenomenon called time decay, and falls to zero exactly at expiration when the option's value equals only its intrinsic value.

#407 · Derivatives

What is the payoff formula for a long call option at expiration?

Answer

$\text{Max}(0, \text{spot price at expiration} - \text{exercise price})$. The holder exercises only if the spot price exceeds the exercise price.

#408 · Derivatives

What is the payoff formula for a long put option at expiration?

Answer

$\text{Max}(0, \text{exercise price} - \text{spot price at expiration})$. The holder exercises only if the exercise price exceeds the spot price.

#409 · Derivatives

What is the cost of carry model for forward pricing?

Answer

The forward price equals the spot price compounded at the risk free rate over the contract term, adjusted for carrying costs such as storage and insurance, and reduced by any benefits received from holding the asset, such as dividends or convenience yield.

#410 · Derivatives

What is arbitrage in the context of derivatives pricing?

Answer

A risk free profit opportunity earned by simultaneously exploiting a price discrepancy, such as buying an underpriced asset or contract and selling an equivalent overpriced position, with no net investment and no risk.

#411 · Derivatives

What is cash and carry arbitrage?

Answer

A strategy used when a forward contract is overpriced relative to fair value: the arbitrageur borrows money to buy the underlying asset spot, carries it to expiration, and simultaneously sells the overpriced forward, locking in a risk free profit.

#412 · Derivatives

What is reverse cash and carry arbitrage?

Answer

A strategy used when a forward contract is underpriced relative to fair value: the arbitrageur sells the underlying asset short, invests the proceeds at the risk free rate, and buys the underpriced forward, locking in a risk free profit.

#413 · Derivatives

What is the law of one price?

Answer

Two assets or portfolios that produce identical future cash flows under all future states of the world must trade at the same current price, or an arbitrage opportunity would exist.

#414 · Derivatives

What does it mean to replicate a derivative payoff?

Answer

Constructing a portfolio of other assets, such as the underlying asset combined with risk free borrowing or lending, whose cash flows exactly match the derivative's payoff, allowing the derivative to be valued off the cost of that portfolio.

#415 · Derivatives

State put-call parity for European options.

Answer

Call price plus present value of the exercise price equals put price plus underlying spot price, for European options with the same exercise price and expiration. It links call and put values through a no-arbitrage relationship.

#416 · Derivatives

How can put-call parity be used to create a synthetic call option?

Answer

Rearranging put-call parity shows that a long call can be replicated by holding a long put, a long position in the underlying asset, and borrowing the present value of the exercise price.

#417 · Derivatives

What is convenience yield?

Answer

A non-monetary benefit of physically holding a commodity, such as avoiding stockouts or maintaining production flexibility, that lowers the forward price relative to a pure cost of carry model because it acts like a return earned by the holder.

#418 · Derivatives

Why do futures contracts have essentially no counterparty credit risk compared with forwards?

Answer

A clearinghouse becomes the counterparty to every futures trade, requires margin deposits, and marks positions to market daily, settling gains and losses each day so that credit exposure never accumulates significantly.

#419 · Derivatives

What is the difference between exchange traded and OTC derivatives in terms of standardization?

Answer

Exchange traded derivatives, like futures and listed options, have standardized contract sizes, expiration dates, and terms set by the exchange. OTC derivatives, like most forwards and swaps, are privately negotiated and customized between counterparties.

#420 · Derivatives

How does a fiduciary call relate to a protective put under put-call parity?

Answer

A fiduciary call, a long call plus a risk free bond paying the exercise price at expiration, has the same payoff and cost as a protective put, a long position in the underlying plus a long put, illustrating the equivalence at the heart of put-call parity.

#421 · Alternative Investments

What are the seven main categories of alternative investments covered in the CFA curriculum?

Answer

Hedge funds, private equity, private debt, real estate, infrastructure, commodities, and digital assets.

#422 · Alternative Investments

What is the typical fee structure for a hedge fund?

Answer

A management fee plus a performance fee, commonly expressed as 2 and 20, meaning 2 percent of assets under management annually plus 20 percent of profits above a benchmark or hurdle.

#423 · Alternative Investments

What is a hurdle rate in a hedge fund fee structure?

Answer

The minimum return the fund must achieve before the manager can collect a performance fee, protecting investors from paying incentive fees on mediocre returns.

#424 · Alternative Investments

What is a high water mark?

Answer

A provision requiring a hedge fund manager to recover any prior losses before collecting performance fees again, preventing fees on gains that merely offset earlier losses.

#425 · Alternative Investments

Define a hard hurdle rate versus a soft hurdle rate.

Answer

With a hard hurdle, the manager earns performance fees only on returns above the hurdle. With a soft hurdle, once the hurdle is cleared the manager earns fees on the entire return, not just the excess.

#426 · Alternative Investments

What is an equity market neutral hedge fund strategy?

Answer

A strategy that takes offsetting long and short positions in equities to eliminate net market exposure, aiming to profit from relative mispricing rather than overall market direction.

#427 · Alternative Investments

What is convertible arbitrage?

Answer

A strategy that goes long a convertible bond while shorting the underlying stock, seeking to profit from pricing inefficiencies between the convertible and the equity while hedging market risk.

#428 · Alternative Investments

What is fixed income arbitrage?

Answer

A strategy exploiting small pricing discrepancies between related fixed income securities, often using leverage to magnify modest yield spreads into attractive returns.

#429 · Alternative Investments

What is merger arbitrage (risk arbitrage)?

Answer

A strategy that buys the target company's stock and often shorts the acquirer's stock in an announced merger, profiting from the spread between the current price and the deal price if the merger closes.

#430 · Alternative Investments

What is a global macro hedge fund strategy?

Answer

A strategy that takes positions across currencies, interest rates, commodities, and equities based on top down analysis of macroeconomic trends and policy shifts.

#431 · Alternative Investments

What distinguishes a fund of hedge funds from a single-manager hedge fund?

Answer

A fund of funds invests in a portfolio of underlying hedge funds, offering diversification and manager selection expertise, but it layers an additional fee on top of the underlying funds' fees.

#432 · Alternative Investments

What is a lockup period in a hedge fund?

Answer

A minimum time after investing during which an investor cannot withdraw capital, allowing the manager to pursue less liquid strategies without facing redemption pressure.

#433 · Alternative Investments

What is a gate provision in a hedge fund?

Answer

A restriction limiting the amount of capital that can be redeemed at any one time, protecting remaining investors from forced asset sales during periods of heavy redemptions.

#434 · Alternative Investments

What is private equity?

Answer

Equity capital invested in companies that are not publicly traded, typically through buyouts of mature firms or growth equity investments in expanding companies, usually held in illiquid, long-term structures.

#435 · Alternative Investments

What is a leveraged buyout (LBO)?

Answer

An acquisition of a company financed with a significant amount of debt, using the target company's own assets and cash flows as collateral and repayment source, with the private equity sponsor providing the remaining equity.

#436 · Alternative Investments

What is venture capital?

Answer

Private equity financing provided to early stage, high growth potential companies in exchange for equity, typically across seed, early, and later stage rounds.

#437 · Alternative Investments

Describe the typical private equity fund structure.

Answer

A limited partnership in which the general partner manages the fund and makes investment decisions, and limited partners contribute capital and have limited liability, usually with a 10 year fund life.

#438 · Alternative Investments

What is a capital call in private equity?

Answer

A request from the general partner for limited partners to fund a portion of their committed capital, drawn down over time as the fund identifies and closes investments.

#439 · Alternative Investments

What is carried interest?

Answer

The share of a private equity fund's profits, typically around 20 percent, paid to the general partner as performance compensation after limited partners receive their capital back and any preferred return.

#440 · Alternative Investments

What is the J-curve effect in private equity?

Answer

The pattern where a fund shows negative returns in early years due to fees and initial investment costs, before returns turn positive as portfolio companies mature and are exited.

#441 · Alternative Investments

What are common private equity exit strategies?

Answer

An initial public offering, a trade sale to a strategic buyer, a secondary sale to another private equity firm, or a recapitalization.

#442 · Alternative Investments

What is private debt (private credit)?

Answer

Debt financing provided by non-bank lenders, such as private credit funds, directly to companies, often filling a gap left by reduced bank lending to middle market or higher risk borrowers.

#443 · Alternative Investments

What is direct lending in private debt?

Answer

A private debt strategy in which a fund originates loans directly to companies rather than purchasing existing loans in the secondary market, often earning an illiquidity premium over comparable public debt.

#444 · Alternative Investments

What is distressed debt investing?

Answer

A strategy that buys the debt of financially troubled companies at a discount, seeking profit from a successful restructuring, recovery, or by gaining control of the company through the reorganization process.

#445 · Alternative Investments

What is mezzanine debt?

Answer

A hybrid form of financing that ranks below senior debt but above equity in the capital structure, often including equity features like warrants, and carrying higher risk and return than senior loans.

#446 · Alternative Investments

What are the two broad approaches to real estate investment?

Answer

Direct ownership of physical property and indirect investment through vehicles such as real estate investment trusts (REITs), limited partnerships, or commingled real estate funds.

#447 · Alternative Investments

What is a REIT?

Answer

A real estate investment trust, a publicly or privately traded company that owns, operates, or finances income producing real estate and typically must distribute most of its taxable income to shareholders.

#448 · Alternative Investments

What are the main return sources for real estate investment?

Answer

Income from rents and property operations, plus capital appreciation from an increase in property value over the holding period.

#449 · Alternative Investments

What is the income approach to real estate valuation?

Answer

A method that estimates property value by capitalizing expected net operating income using a capitalization rate derived from comparable property sales.

#450 · Alternative Investments

What is net operating income (NOI) in real estate?

Answer

Rental and other property income minus operating expenses, excluding financing costs and capital expenditures, used as the basis for income based valuation approaches.

#451 · Alternative Investments

What is infrastructure investing?

Answer

Investment in long lived physical assets that provide essential services, such as toll roads, airports, utilities, and pipelines, typically offering stable, inflation linked cash flows over long horizons.

#452 · Alternative Investments

Distinguish greenfield from brownfield infrastructure investments.

Answer

Greenfield investments fund the construction of new infrastructure assets and carry higher development and completion risk, while brownfield investments acquire existing, already operating assets with more predictable cash flows.

#453 · Alternative Investments

Why do institutional investors find infrastructure assets attractive?

Answer

They tend to have stable, often regulated or contracted cash flows, low correlation with traditional asset classes, and inflation linked revenue, matching well against long dated liabilities.

#454 · Alternative Investments

How can an investor gain exposure to commodities?

Answer

Through direct physical ownership, commodity futures contracts, commodity linked exchange traded funds, or shares of commodity producing companies, with futures being the most common approach for institutional investors.

#455 · Alternative Investments

What is contango in a commodity futures market?

Answer

A market condition in which futures prices are higher than the expected spot price, so the futures curve slopes upward, typically producing a negative roll yield for a long futures position.

#456 · Alternative Investments

What is backwardation in a commodity futures market?

Answer

A market condition in which futures prices are lower than the expected spot price, so the futures curve slopes downward, typically producing a positive roll yield for a long futures position.

#457 · Alternative Investments

What is roll yield in commodity futures investing?

Answer

The return earned or lost from selling an expiring futures contract and buying a longer dated one, positive when the market is in backwardation and negative when the market is in contango.

#458 · Alternative Investments

What are cryptocurrencies as an alternative investment?

Answer

Digital assets that use cryptography and distributed ledger technology to record and verify transactions, characterized by high volatility, no cash flows, and valuation based largely on supply, demand, and network adoption rather than fundamentals.

#459 · Alternative Investments

What role does blockchain play in digital assets?

Answer

Blockchain is the underlying distributed ledger technology that records transactions across a network of computers in a way intended to be transparent, secure, and resistant to alteration, forming the infrastructure for cryptocurrencies and other digital assets.

#460 · Alternative Investments

What are common risks specific to alternative investments as an asset class?

Answer

Illiquidity, limited transparency and less regulatory oversight, valuation difficulty for infrequently traded assets, use of leverage, and generally higher fees than traditional investments.

#461 · Portfolio Management

What are the three steps of the portfolio management process?

Answer

Planning (assess needs and constraints, write the IPS), execution (asset allocation, security analysis and selection, portfolio construction), and feedback (monitor and rebalance the portfolio, evaluate performance against the IPS objectives).

#462 · Portfolio Management

What core idea does modern portfolio theory add to the portfolio management process?

Answer

It shows that combining assets with imperfect correlation reduces overall portfolio risk without proportionally reducing expected return, so a portfolio should be evaluated as a whole rather than asset by asset.

#463 · Portfolio Management

Why do individuals hold portfolios instead of single securities?

Answer

A single security exposes an investor to firm-specific risk that diversification can remove for free, so a diversified portfolio offers a better risk-return tradeoff than concentrated holdings of the same expected return.

#464 · Portfolio Management

List the main types of institutional investors covered in CFA Level I.

Answer

Defined benefit pension plans, defined contribution pension plans, endowments and foundations, banks, insurance companies (life and non-life), investment companies, and sovereign wealth funds.

#465 · Portfolio Management

How does a defined benefit (DB) pension plan differ from a defined contribution (DC) plan in risk bearing?

Answer

In a DB plan the sponsor bears the investment risk and promises a specified retirement benefit; in a DC plan the employee bears the investment risk and the retirement benefit depends on account performance.

#466 · Portfolio Management

What distinguishes a life insurance company's investment needs from a non-life (property and casualty) insurer's?

Answer

Life insurers generally have longer, more predictable liability horizons and can accept more illiquidity and longer duration; non-life insurers face shorter and less predictable claim timing, so they need greater liquidity and shorter duration assets.

#467 · Portfolio Management

What is the typical time horizon and liquidity need of a university endowment?

Answer

Endowments generally have a very long or perpetual time horizon and support a fixed annual spending rule, so they need moderate liquidity for spending but can tolerate significant illiquidity in the remainder of the portfolio.

#468 · Portfolio Management

What is an Investment Policy Statement (IPS)?

Answer

A written planning document that states an investor's return objectives and risk tolerance along with all constraints, and that guides investment decisions and serves as the benchmark for evaluating portfolio performance.

#469 · Portfolio Management

What are the two main objectives specified in an IPS?

Answer

Risk objectives (how much risk the investor is able and willing to bear, stated as ability and willingness to take risk) and return objectives (a required or desired return, stated on an absolute or relative basis).

#470 · Portfolio Management

List the five standard IPS constraint categories.

Answer

Liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances.

#471 · Portfolio Management

Distinguish ability to take risk from willingness to take risk in an IPS.

Answer

Ability to take risk is an objective measure based on financial factors such as wealth, income stability, and time horizon; willingness to take risk is a subjective, psychological measure of the investor's comfort with volatility. When they conflict, the more conservative of the two generally governs.

#472 · Portfolio Management

What does the liquidity constraint in an IPS address?

Answer

The need to convert assets to cash quickly without material loss of value to meet anticipated and unanticipated spending needs, such as planned withdrawals, an emergency reserve, or upcoming known cash outflows.

#473 · Portfolio Management

Give an example of a legal and regulatory constraint in an IPS.

Answer

Prudent investor rules governing a trustee's duties, restrictions on pension fund investments under regulation such as ERISA, or insider trading rules that restrict an executive's ability to trade their own company's stock.

#474 · Portfolio Management

What is meant by systematic risk?

Answer

Risk that affects the entire market or economy, such as interest rate, inflation, or business cycle risk, that cannot be eliminated through diversification and is the only risk compensated with a risk premium.

#475 · Portfolio Management

What is meant by unsystematic (idiosyncratic) risk?

Answer

Firm-specific or industry-specific risk, such as a lawsuit, management change, or product recall, that can be reduced or eliminated by holding a well-diversified portfolio and therefore earns no risk premium.

#476 · Portfolio Management

Roughly how many randomly selected stocks are needed to eliminate most unsystematic risk in a portfolio?

Answer

Empirical studies show that around 30 or more randomly selected stocks eliminate most diversifiable risk, though some systematic risk always remains regardless of the number of holdings.

#477 · Portfolio Management

What is the formula for the expected return of a two-asset portfolio?

Answer

$E(R_p) = w_1 \times E(R_1) + w_2 \times E(R_2)$, where w_1 and w_2 are the portfolio weights of assets 1 and 2 and $w_1 + w_2 = 100\%$.

#478 · Portfolio Management

What is the formula for the variance of a two-asset portfolio?

Answer

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \text{Cov}(R_1, R_2)$, where $\text{Cov}(R_1, R_2) = \rho_{1,2} \times \sigma_1 \times \sigma_2$ and $\rho_{1,2}$ is the correlation between the two assets' returns.

#479 · Portfolio Management

How does the correlation coefficient between two assets affect diversification benefit?

Answer

The lower the correlation coefficient (closer to negative 1), the greater the risk reduction from combining the assets; a correlation of positive 1 provides no diversification benefit, since portfolio risk is then just the weighted average of the individual risks.

#480 · Portfolio Management

What is the minimum-variance frontier?

Answer

The set of portfolios that has the minimum variance (lowest risk) for each level of expected return, plotted with standard deviation on the x-axis and expected return on the y-axis.

#481 · Portfolio Management

What is the efficient frontier?

Answer

The upper portion of the minimum-variance frontier starting at the global minimum-variance portfolio, containing only portfolios that offer the highest expected return for each level of risk (no other portfolio dominates them).

#482 · Portfolio Management

What defines an investor's optimal portfolio on the efficient frontier?

Answer

The point where the investor's highest-attainable indifference curve (representing their risk aversion and utility tradeoff between risk and return) is tangent to the efficient frontier.

#483 · Portfolio Management

What is the Capital Allocation Line (CAL)?

Answer

A line that shows the risk-return combinations available by combining a risky portfolio with the risk-free asset, plotted from the risk-free rate through the risky portfolio in expected return versus standard deviation space.

#484 · Portfolio Management

What is the formula for the slope of the Capital Allocation Line?

Answer

Slope = $[E(R_p) \text{ minus } R_f] \div \sigma_p$, which is the Sharpe ratio of the risky portfolio, where $E(R_p)$ is the risky portfolio's expected return, R_f is the risk-free rate, and σ_p is the risky portfolio's standard deviation.

#485 · Portfolio Management

What is the Capital Market Line (CML)?

Answer

The specific Capital Allocation Line that results when the risky portfolio is the market portfolio; every point on it represents an efficient combination of the risk-free asset and the market portfolio, and it has the steepest attainable slope for a rational investor.

#486 · Portfolio Management

What two-fund separation theorem underlies the CML?

Answer

All investors, regardless of risk preference, should hold the same optimal risky portfolio (the market portfolio) and adjust their overall risk only by varying the mix between that portfolio and the risk-free asset.

#487 · Portfolio Management

What is systematic risk measured by, and what does it capture?

Answer

Beta, which measures an asset's sensitivity to movements in the overall market portfolio; it captures only the non-diversifiable, market-related portion of an asset's total risk.

#488 · Portfolio Management

What is the formula for beta?

Answer

$\beta_i = \text{Cov}(R_i, R_m) \div \sigma_m^2 = \rho_{i,m} \times (\sigma_i \div \sigma_m)$, where $\text{Cov}(R_i, R_m)$ is the covariance of the asset's return with the market return, σ_m^2 is the market's return variance, and $\rho_{i,m}$ is the correlation between the asset and the market.

#489 · Portfolio Management

What does a beta of 1.0, greater than 1.0, and less than 1.0 mean?

Answer

A beta of 1.0 means the asset moves in line with the market on average; greater than 1.0 means it is more volatile than the market (amplifies market moves); less than 1.0 means it is less volatile than the market (dampens market moves).

#490 · Portfolio Management

What is the Security Market Line (SML)?

Answer

A line, derived from the CAPM, that plots required (expected) return on the y-axis against beta on the x-axis for all assets, whether efficient or not, showing the linear relationship between systematic risk and required return.

#491 · Portfolio Management

How does the SML differ from the CML in what it measures on the x-axis?

Answer

The CML uses total risk (standard deviation) and applies only to efficient portfolios; the SML uses systematic risk (beta) and applies to any individual asset or portfolio, efficient or not.

#492 · Portfolio Management

What is the Capital Asset Pricing Model (CAPM) formula?

Answer

$E(R_i) = R_f + \beta_i \times [E(R_m) \text{ minus } R_f]$, where $E(R_i)$ is the required return on asset i , R_f is the risk-free rate, β_i is the asset's beta, and $E(R_m) \text{ minus } R_f$ is the market risk premium.

#493 · Portfolio Management

What are the four key assumptions underlying the CAPM?

Answer

Investors are risk-averse, utility-maximizing, rational; markets are frictionless with no taxes or transaction costs; investors can borrow and lend unlimited amounts at the risk-free rate; and investors have homogeneous expectations about asset returns, variances, and correlations.

#494 · Portfolio Management

In the CAPM, what is the required return on an asset with a beta of zero?

Answer

It equals the risk-free rate, since the asset has no systematic risk exposure and earns no risk premium above R_f , even if it has significant total (unsystematic) risk.

#495 · Portfolio Management

How is CAPM used to assess whether a security is overvalued or undervalued?

Answer

Compare the security's estimated or forecast return to its CAPM-required return: if the forecast return exceeds the required return the security plots above the SML and is undervalued (buy); if below, it plots below the SML and is overvalued (sell).

#496 · Portfolio Management

What is the market risk premium?

Answer

The market risk premium equals $E(R_m)$ minus R_f , the expected return on the market portfolio minus the risk-free rate; it is the compensation investors require for bearing the systematic risk of the overall market.

#497 · Portfolio Management

What is the beta of the market portfolio and of a risk-free asset?

Answer

The market portfolio has a beta of 1.0 by definition, since it is being compared to itself; a risk-free asset has a beta of 0, since its return does not vary with market movements.

#498 · Portfolio Management

What is the risk-return tradeoff embedded in modern portfolio theory called?

Answer

The efficient frontier concept states that for a given level of risk investors demand the highest possible expected return, and for a given expected return investors prefer the lowest possible risk, which defines efficient portfolios.

#499 · Portfolio Management

Why is total risk (standard deviation) not the correct risk measure for pricing an individual asset held in a diversified portfolio?

Answer

Because diversification eliminates unsystematic risk at no cost, only systematic risk (beta) remains relevant and priced; using total risk would overstate the compensation an investor should require for that asset.

#500 · Portfolio Management

What is the formula for the covariance between two assets given their correlation and standard deviations?

Answer

$\text{Cov}(R_1, R_2) = \rho_{1,2} \times \sigma_1 \times \sigma_2$, where $\rho_{1,2}$ is the correlation coefficient between the two assets' returns and σ_1 and σ_2 are their respective standard deviations.